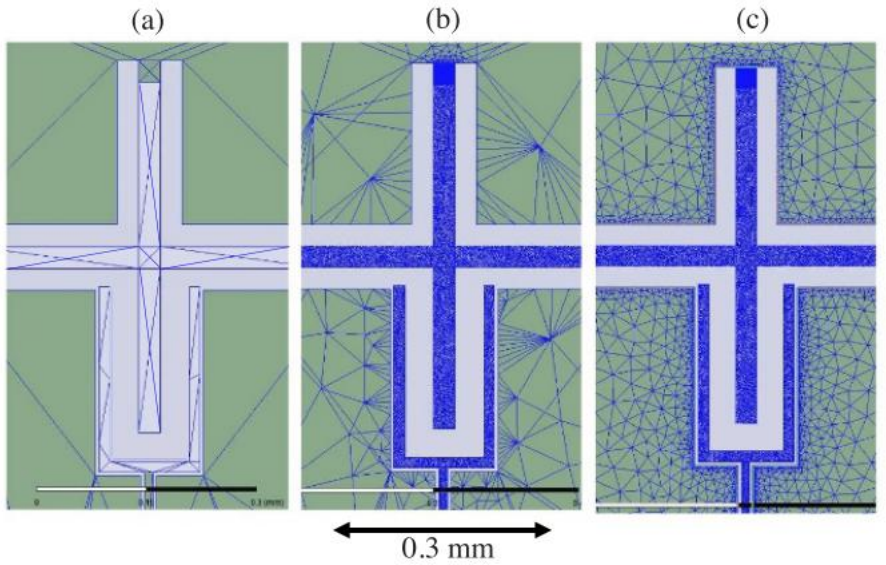
FEM simulation using Ansys HFSS

Editor:謝宗甫

Date:2026/01/10

Why Ansys HFSS?

- ANSYS HFSS/Q3D, a commercial finite element method (FEM) solver and the most commonly used in the community, solves Maxwell's equations by breaking space into a discrete mesh.



Example of mesh after several rounds of refinement

Why simulate?

- **Verify Hamiltonian**: Translation between physical layout and effective Hamiltonian parameters.
- **Identify Parasitics**: Detecting unwanted modes or stray couplings before fabrication.
- **Cost Efficiency**: Reducing expensive cleanroom fabrication iterations.

Which kind of solver?

What we will use here is below:

Eigenmode solver

- Used to find **resonant frequencies(fields) and Q value** of the structure.
- Not Driven Network (Analysis and Composite Excitation are disabled for these solution types.) Eigenmode designs **cannot contain design parameters that depend on frequency**, for example a frequency-dependent impedance boundary condition.

Driven Model solver

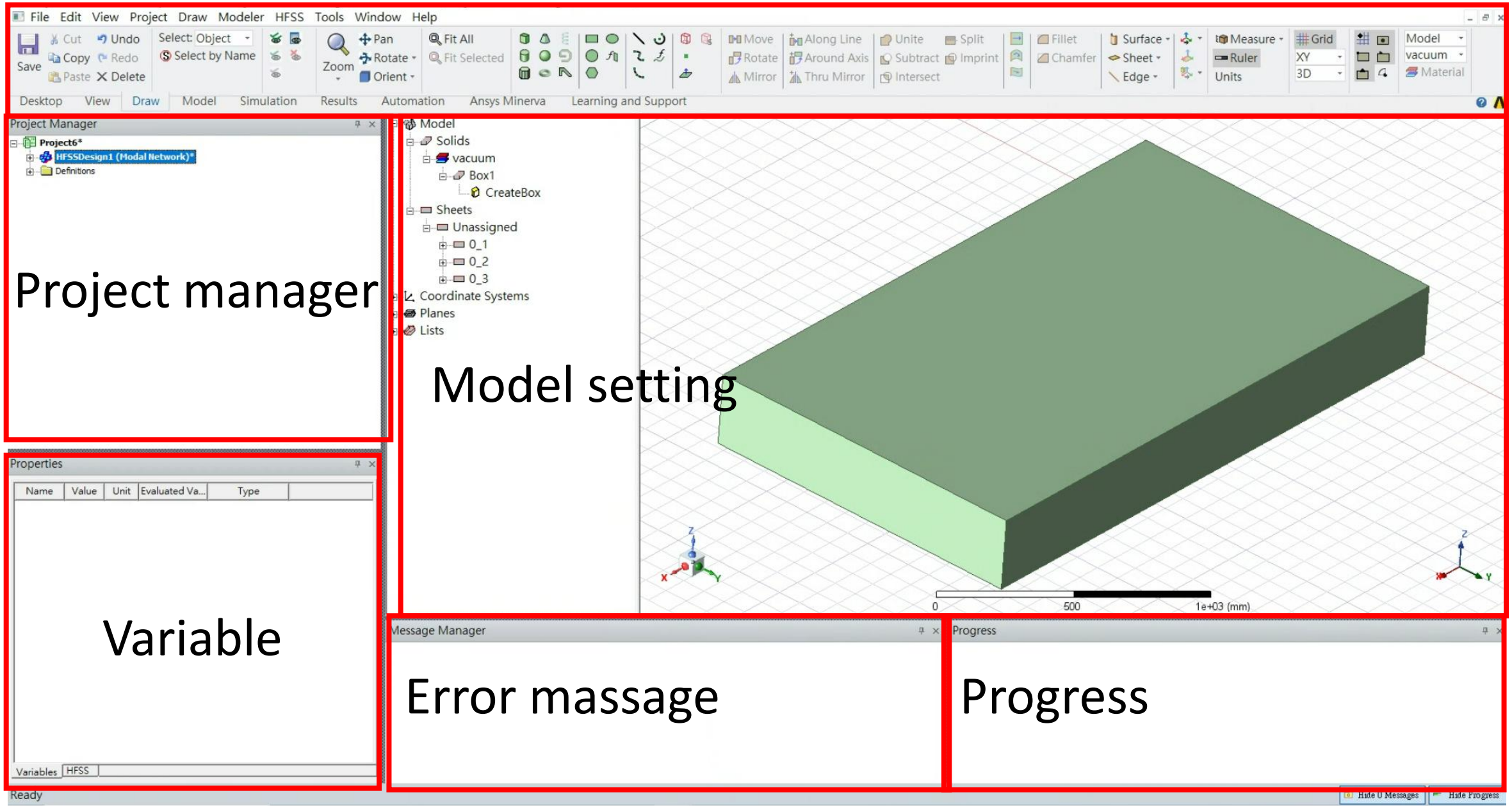
- Used for frequency-dependent scattering parameters (**S-parameters**).
- For calculating the mode-based S-parameters of passive , high-frequency structures such as microstrips, waveguides, and transmission lines which are **"driven“** by a source, and for computing incident plane wave scattering. Network Analysis is the default and functions as before.

Fast review

Feature	**Eigenmode**	**Driven Model**
Goal	Finds natural **Resonant Frequencies** and their associated mode shapes (Q-factors).	Calculates **S-parameter** due to an external source.
Excitation	**None**. No ports or sources are used.	**Required**. Must define Lumped Ports
Outputs	Mode frequencies fr, Quality Factor Q, and Eigenmode fields.	S parameters
Common Use	Find resonant frequencies(decided by Q value)	Antennas, PCBs, connectors, and general high-frequency transmission.

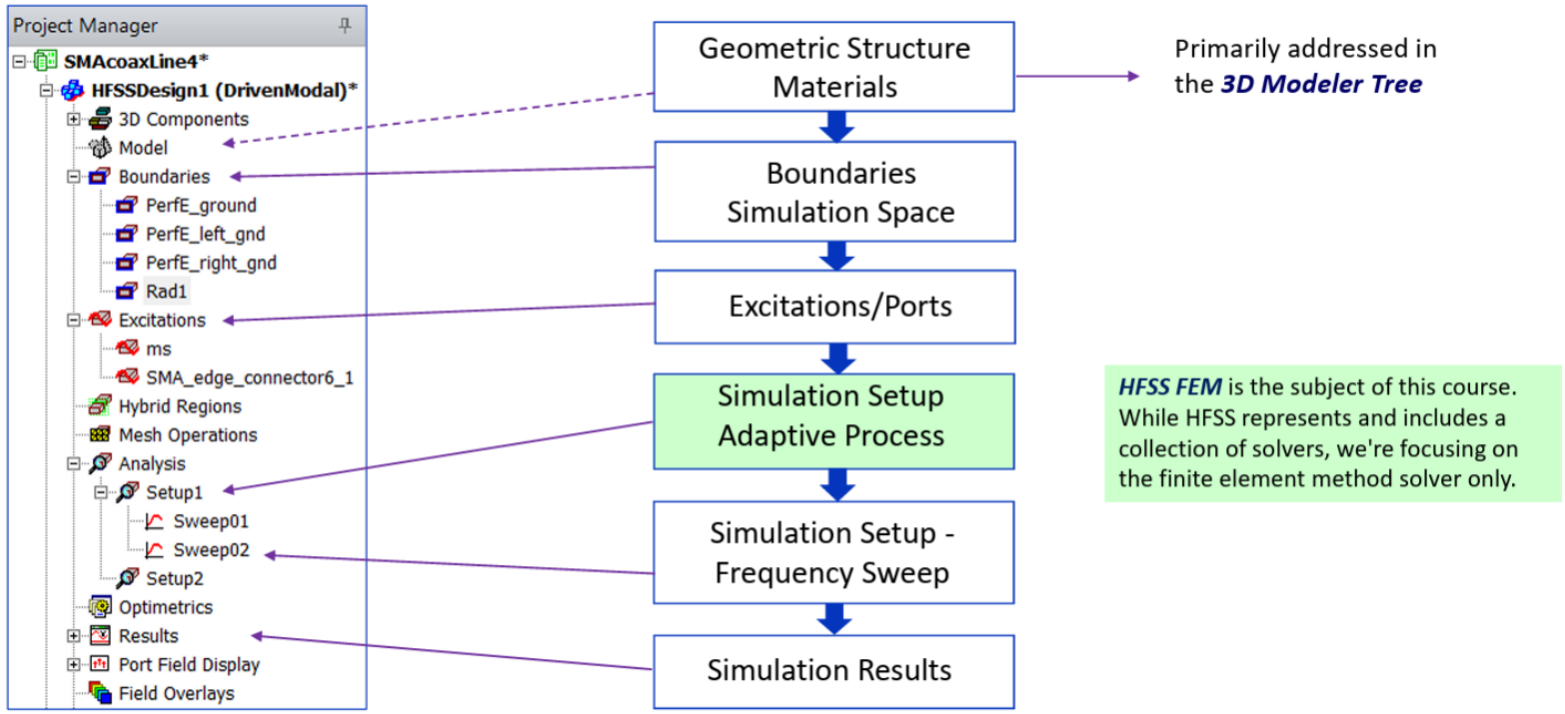
Interface brief

toolbox



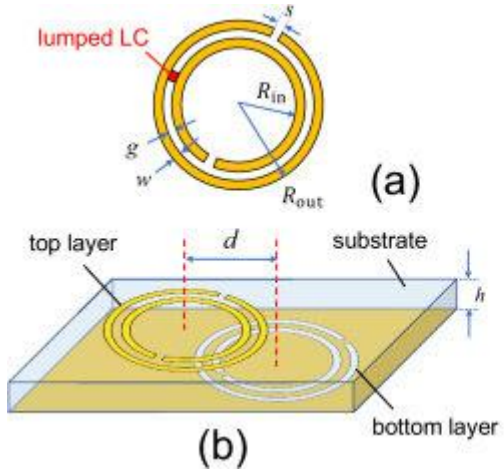
Project manager

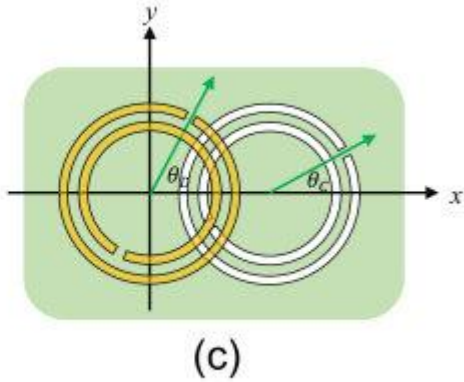
The HFSS Project Manager Reflects EM Simulation Workflow



The document "***An Introduction to HFSS***", Chapter 5 ***"HFSS Modeling GUI Basics"*** section ***"Modeling Practice in HFSS"*** with an HFSS workflow.

SOP for training



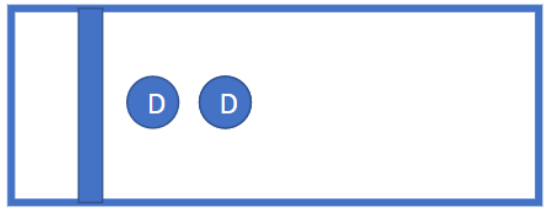


- Example: 2D2C

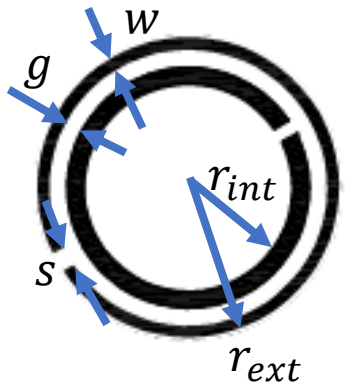
substrate	r_{ext}	r_{int}	w	g	s	d_{in}	d_{out}	l
RO4003C	3.8	3.0	0.4	0.4	0.4	5	10	

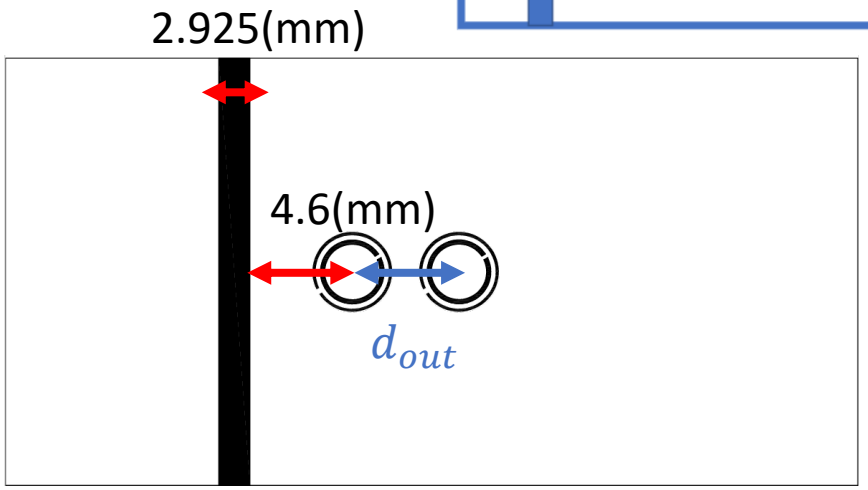
(θ_D, θ_C)
(30°, 60°)

1-port

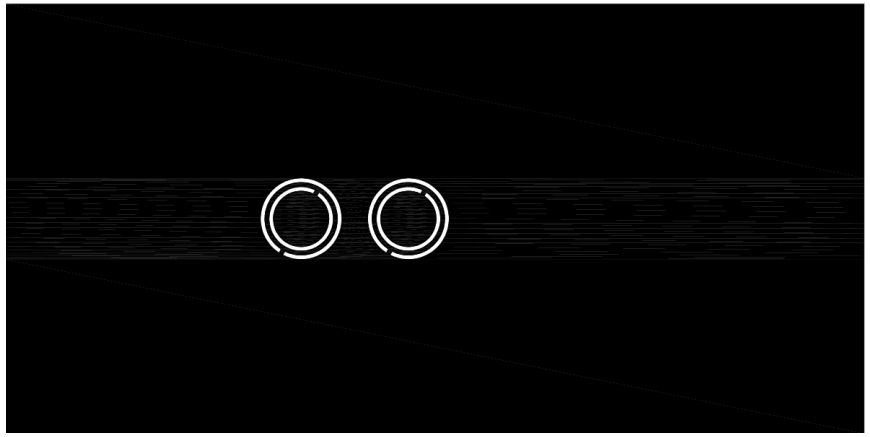








Front DSRR



Back CSRR

(black refers copper)

Brief introduce

- DSRR(Double-sided Split-Ring Resonator)

is a miniaturized metamaterial structure, often consisting of two nested or adjacent split rings, designed to create unique electromagnetic responses



- CSRR(Complementary Split-Ring Resonator)

is a metamaterial element, essentially the "negative" of a standard Split-Ring Resonator, created by removing a ring shape from a metal plane, forming an aperture.

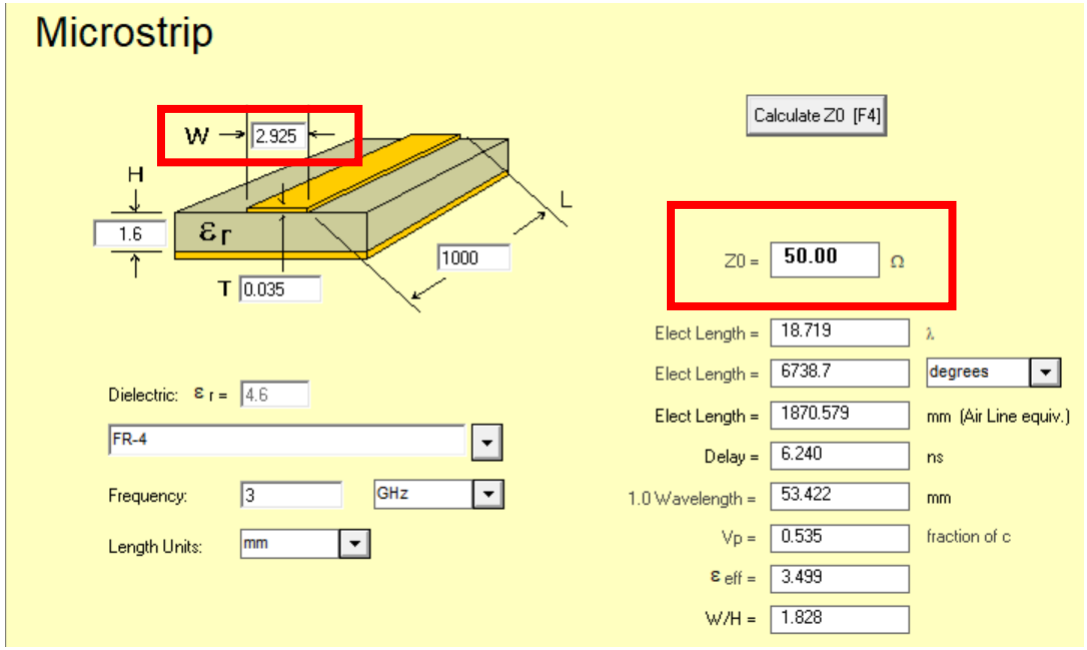


Model set up parameter calculating

- Tool: Appcad
- Click **transmission lines->microstrips**

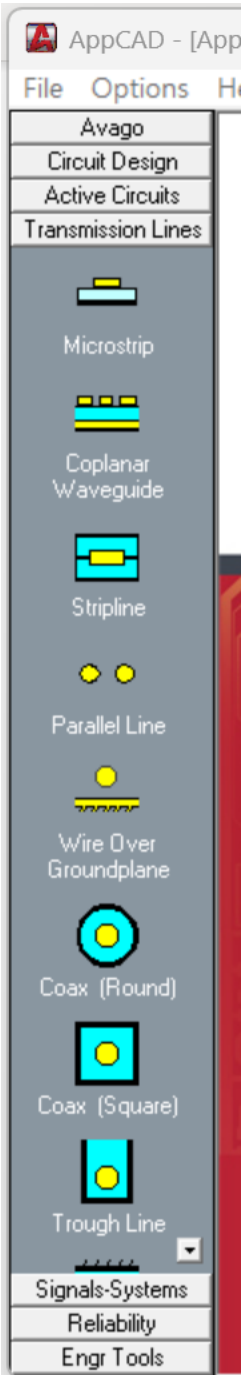
1. Unit set(mm)
2. L=1000
3. Strip thickness T=0.035
4. Substrate thickness H=1.6
5. Freq=3GHz
6. Choose FR-4(PCB material)
7. Calculate Z0 and regulate strip width W till $Z_0 = 50.00\Omega$

Channel width=2.925(mm)

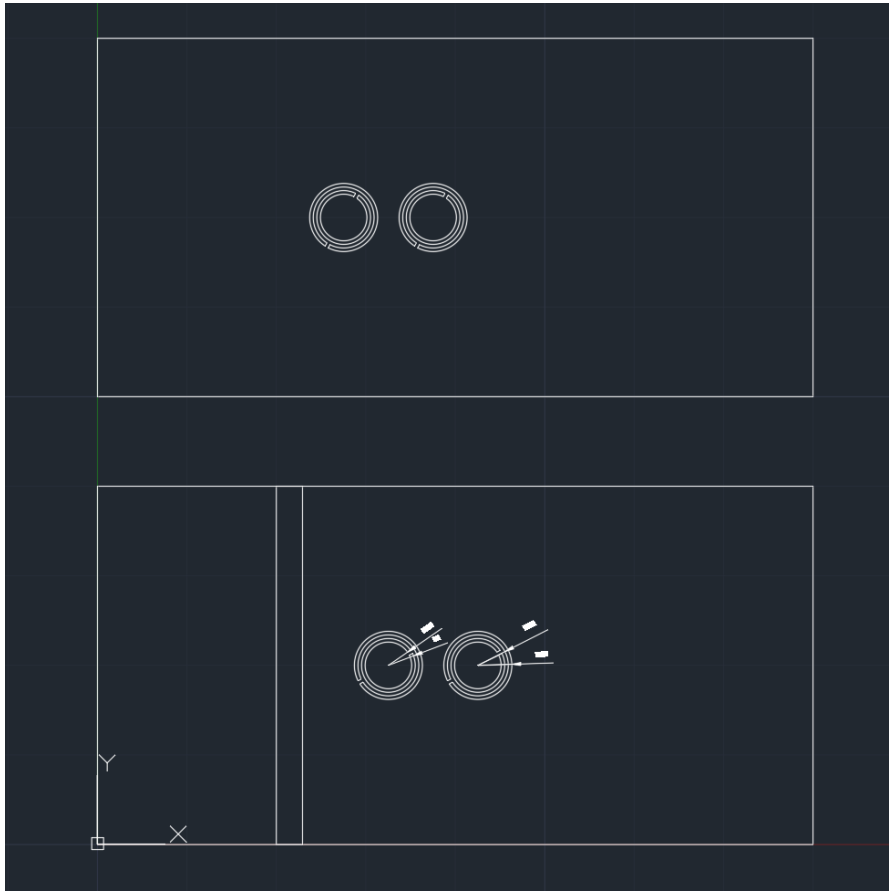


Remember the parameter!!!

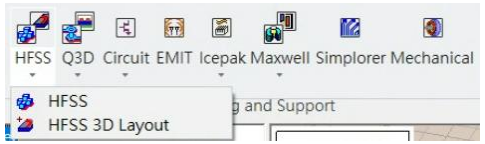
$Gap=, Width=, L=, \lambda=, f=$



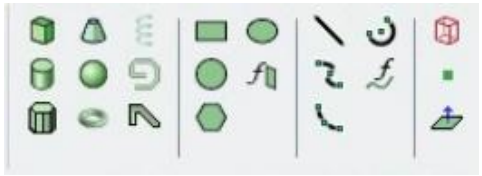
Autocad build up



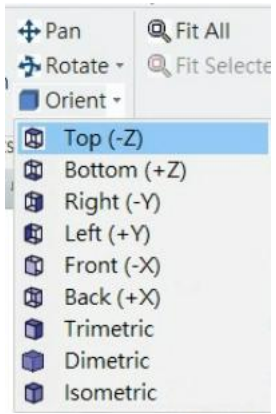
SOP



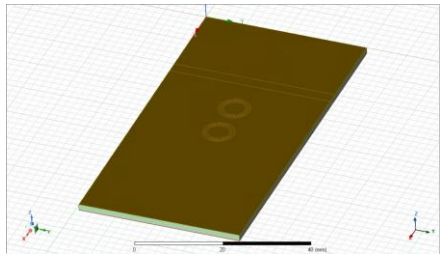
1.Insert HFSS design & solver setting



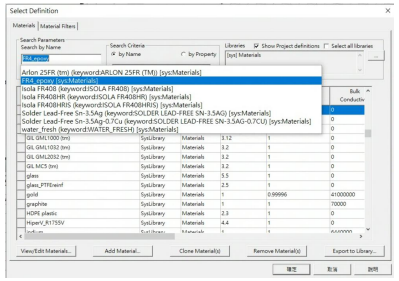
2.Import or draw the model



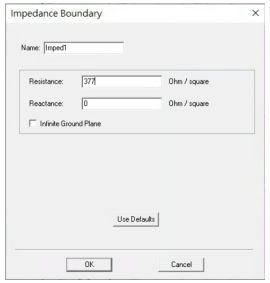
3.Orient setting



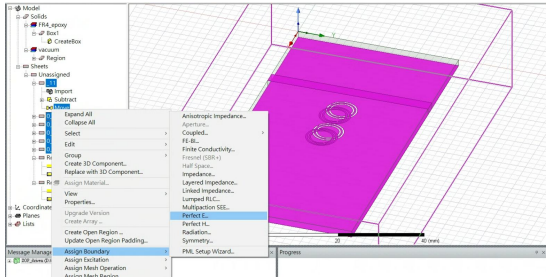
4.Edit model



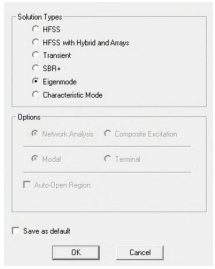
5. Assign material



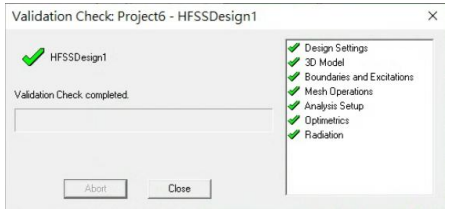
6. Assign boundary



Driven model



Eigenmode

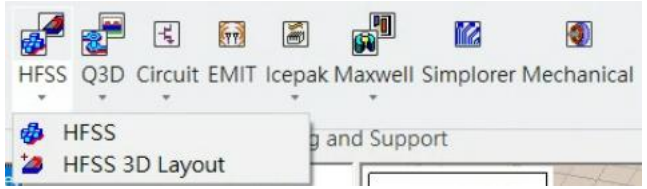


Run simulation

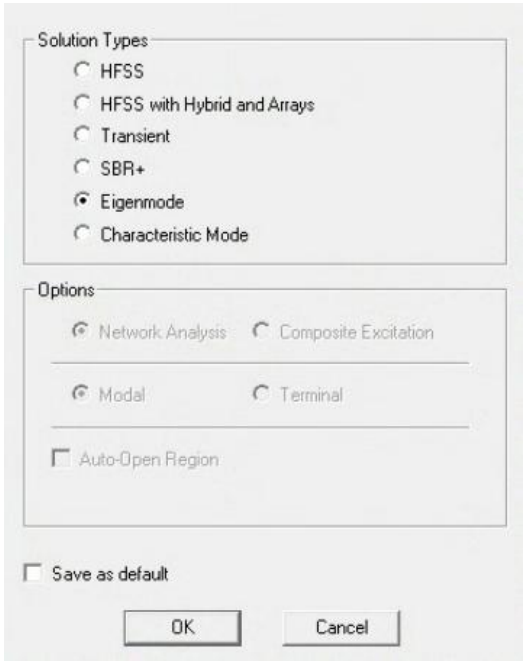
Setting up an HFSS Design

Insert HFSS design

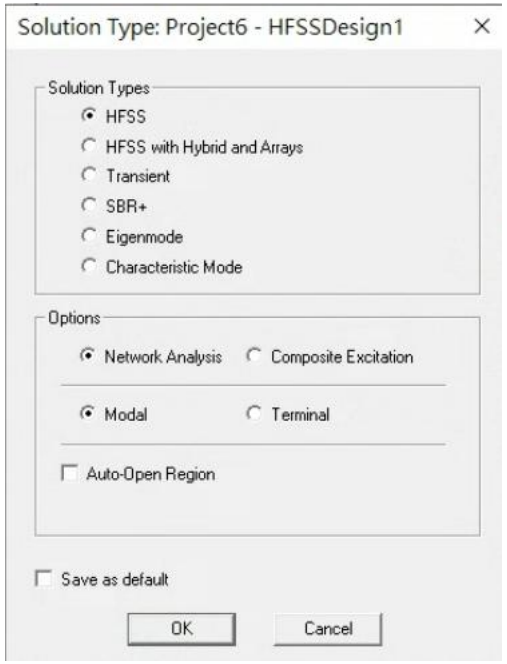
1. Click **HFSS icon to create design**



2. Click **HFSS->solution type**

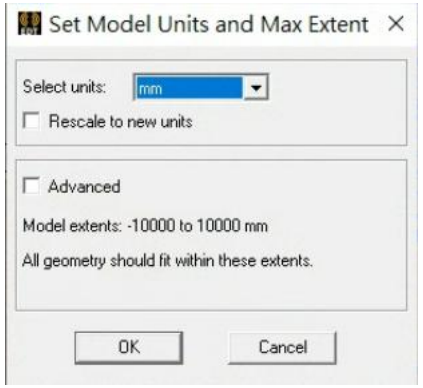


Eigenmode



HFSS and Modal (driven model)

3. Click **modeler->units**



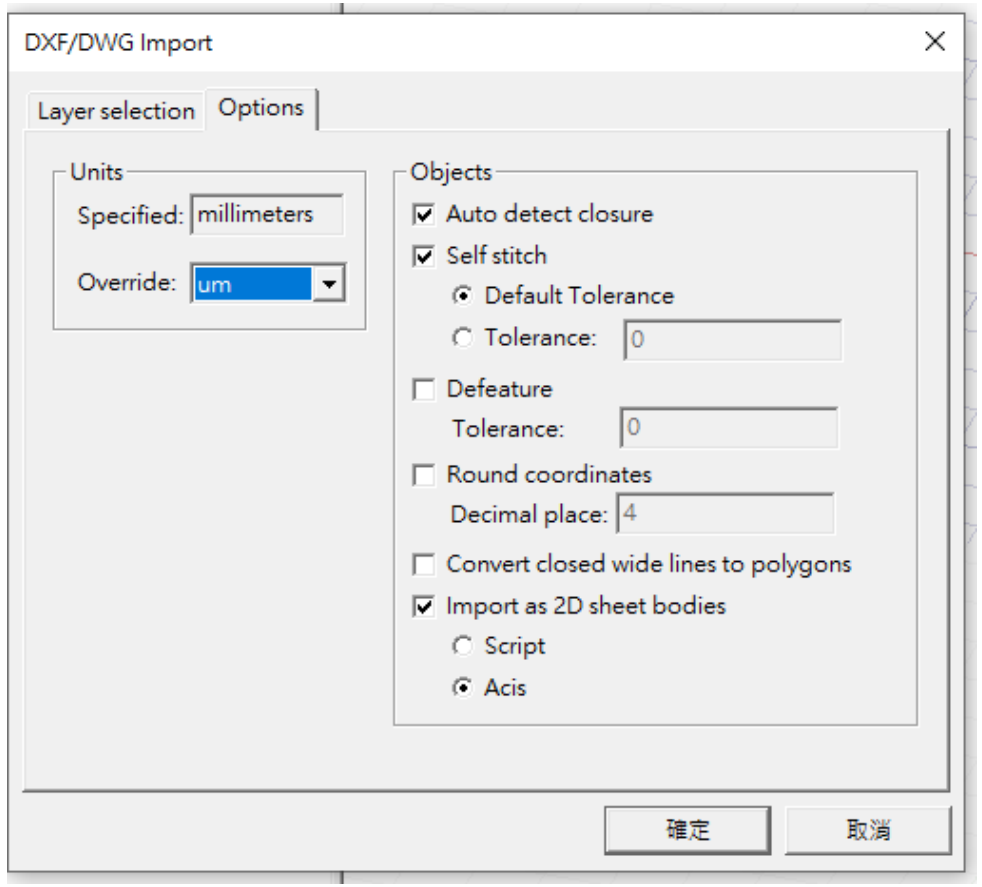
Depend on you (prefer **mm**)

If you have question HFSS help search Selecting the Solution Type

Import or draw the model

1.Click **modeler->import->choose yours->options**

- Override: change the unit to yours(μm)
- ☑ import as 2D sheet bodies



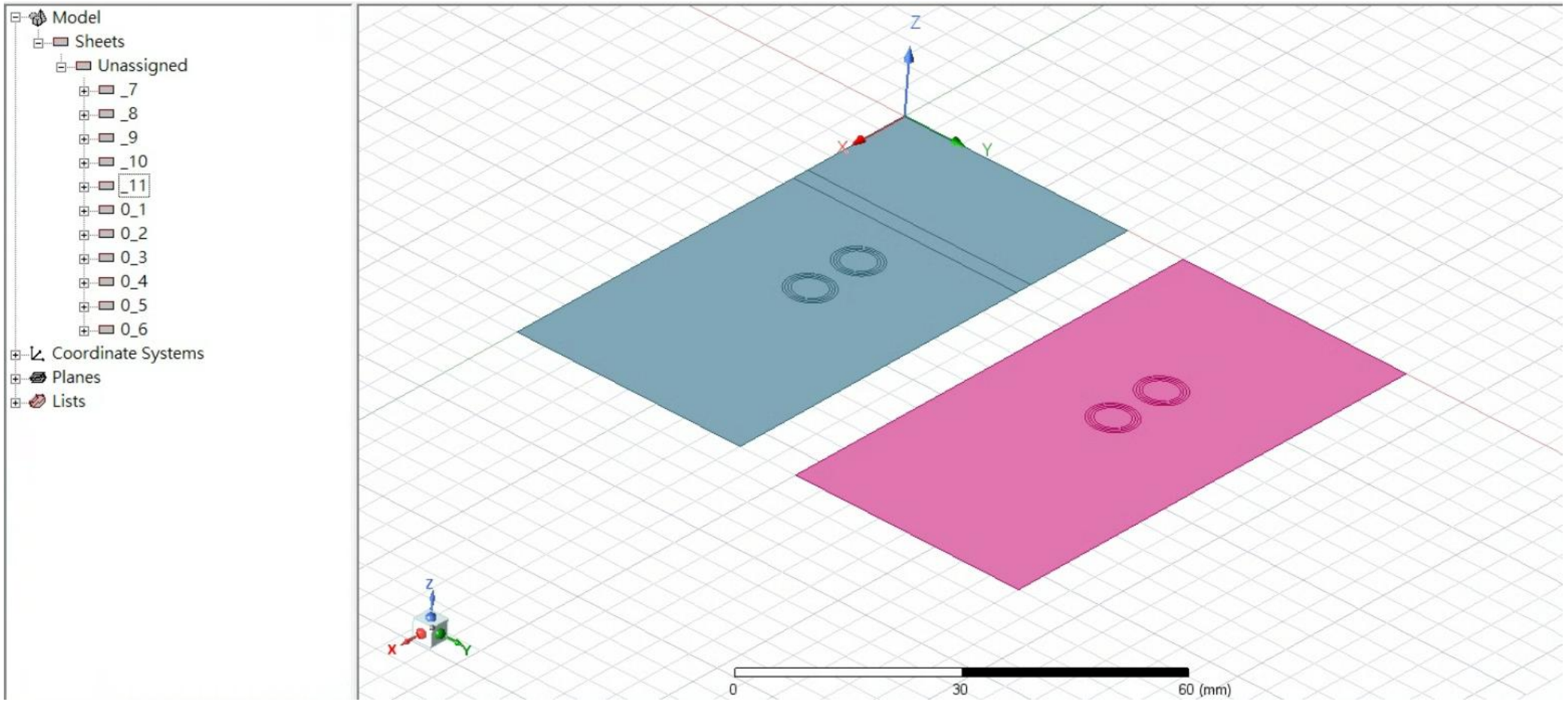
OR draw the model

- Click **draw**



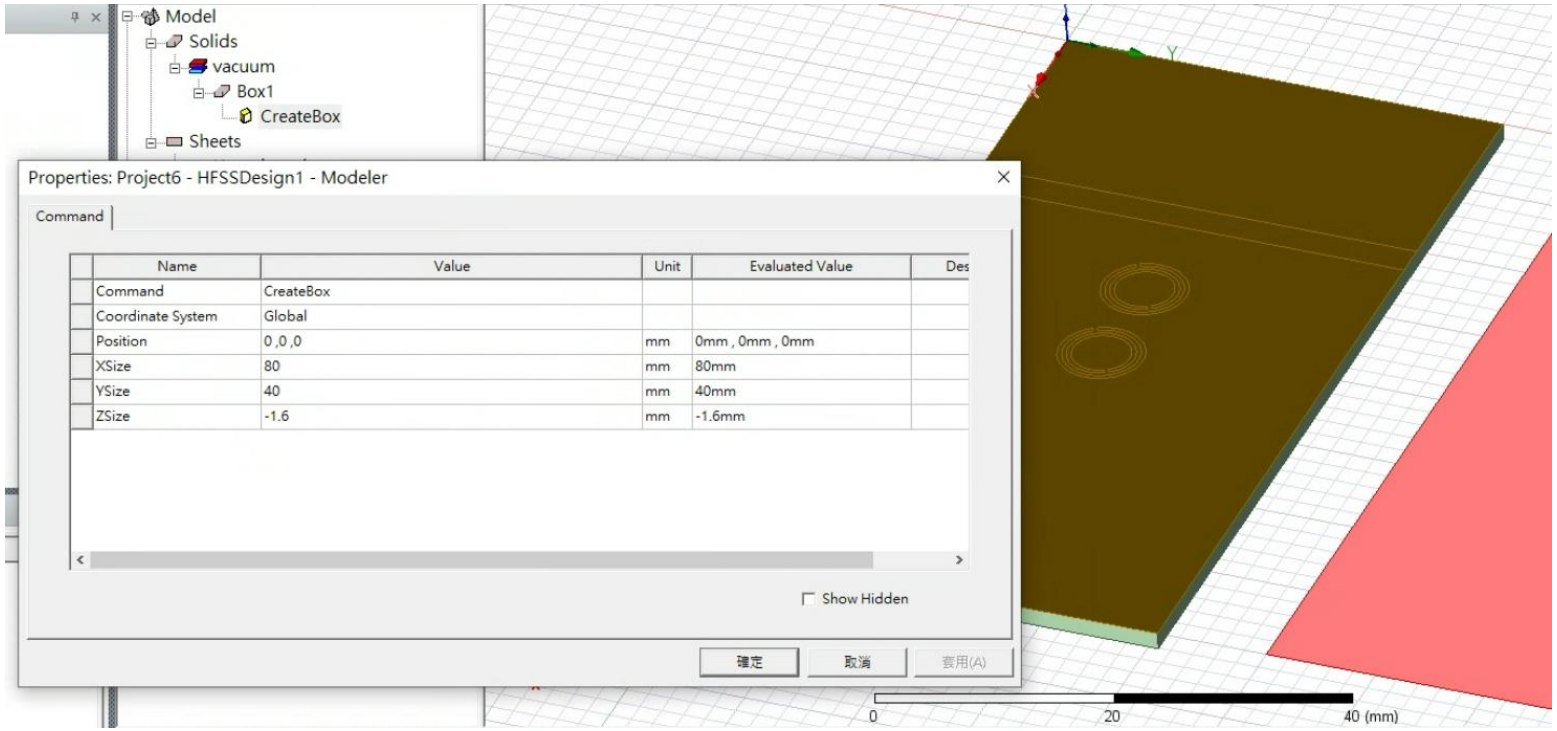
Try to draw each kind of object, familiar with parameter setting

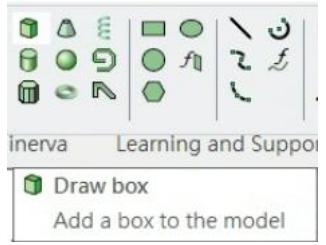
Import result figure



Draw substrate using Box

- **Substrate(PCB)**



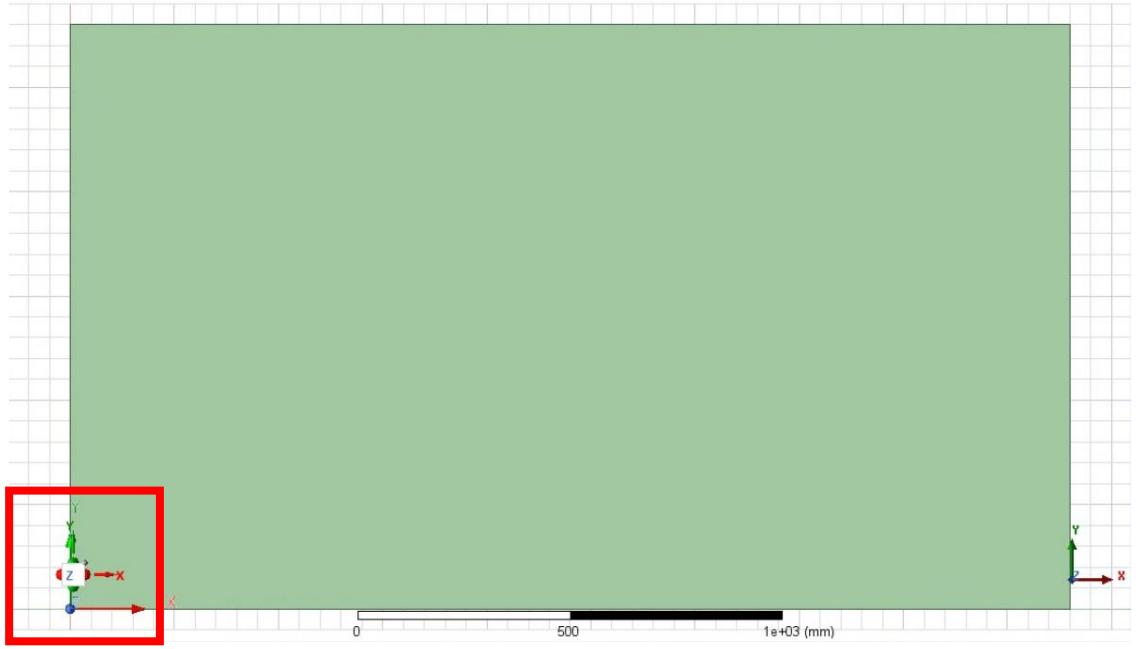


Click draw box icon
->Left click CreateBox icon
xsize=PCB x-size
ysize=PCB y-size
zsize=-1.6(z-axis downward)

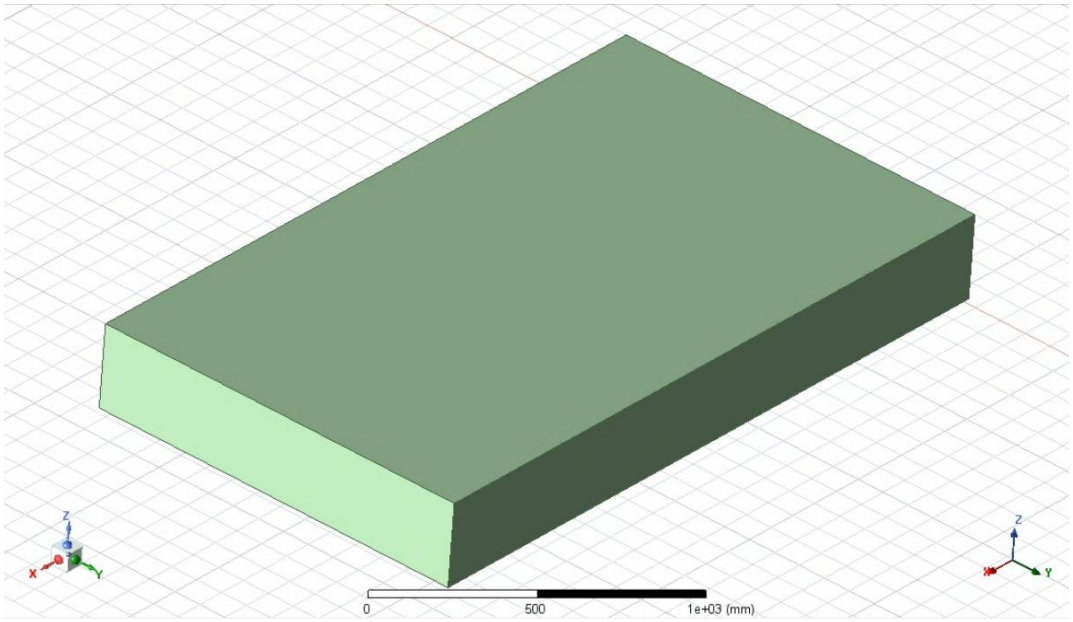


Orient setting(Usual view)

- xy-plane view



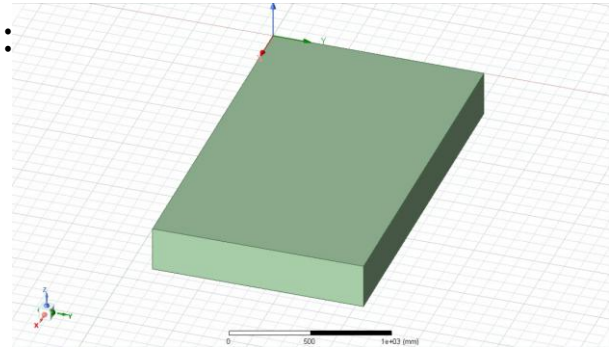
- view2



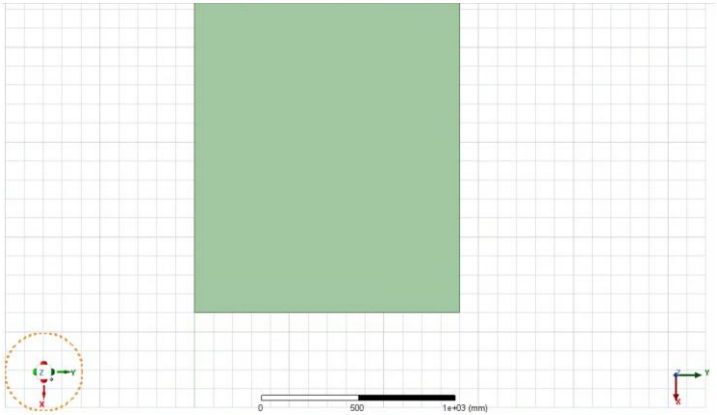
Tips : Use the scroll of the mouse to zoom in or out

Orient setting(xy-plane)

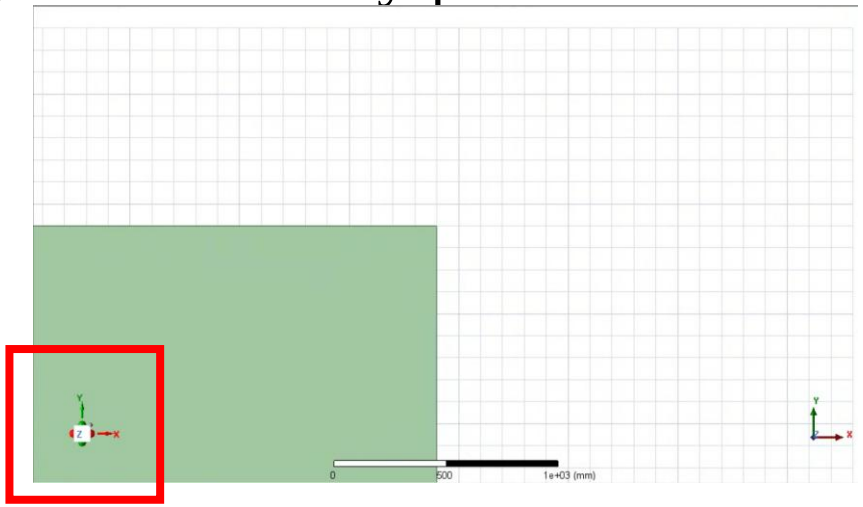
Example:



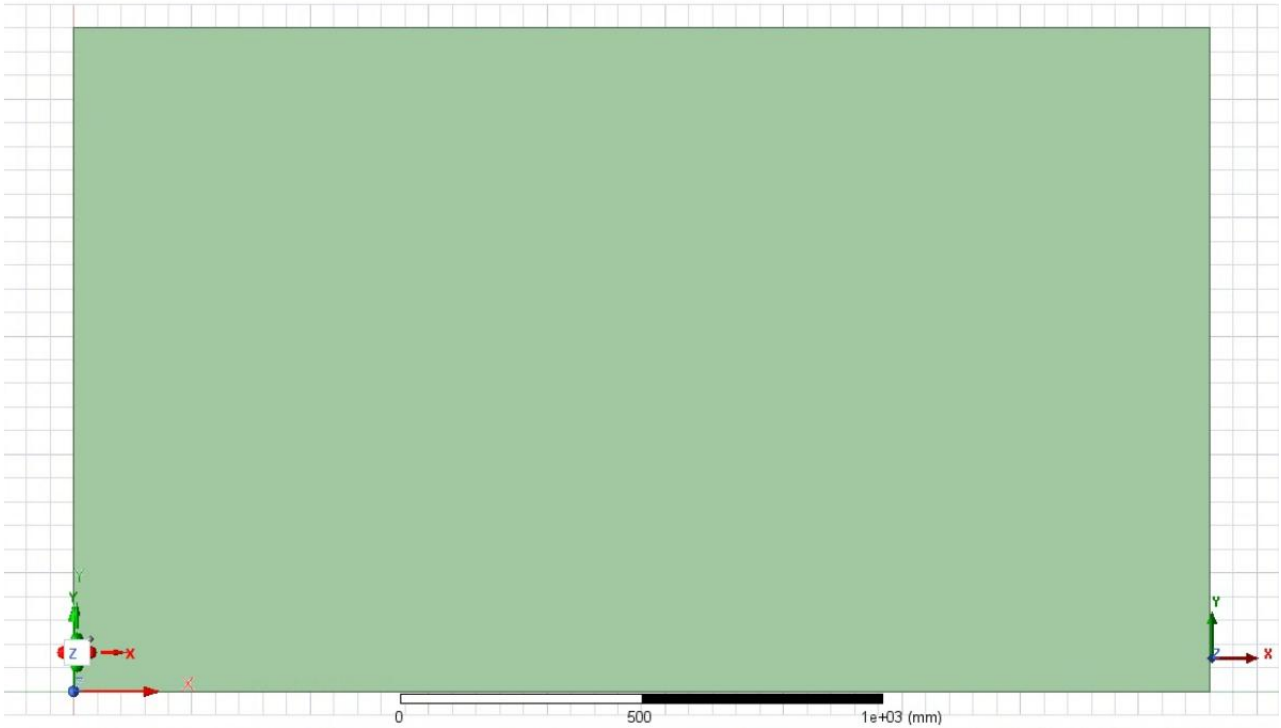
1. Click **draw->orient->top(-z)**



2. Click the **yellow circle** two times/per cycle till usual xy-plane



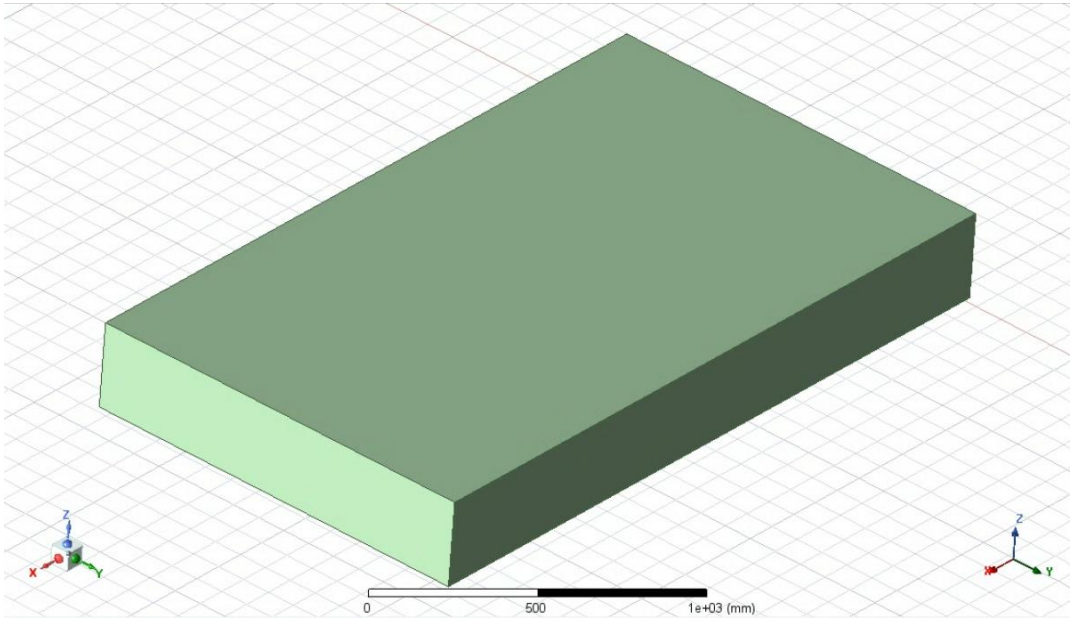
3. **Right click->view->fit all**



Orient setting-view2

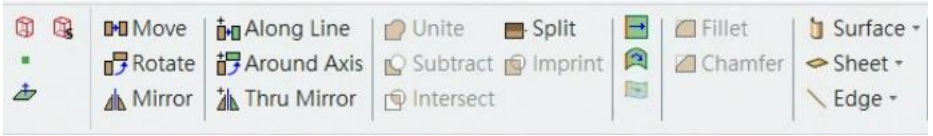
- **Just double click orient icon**



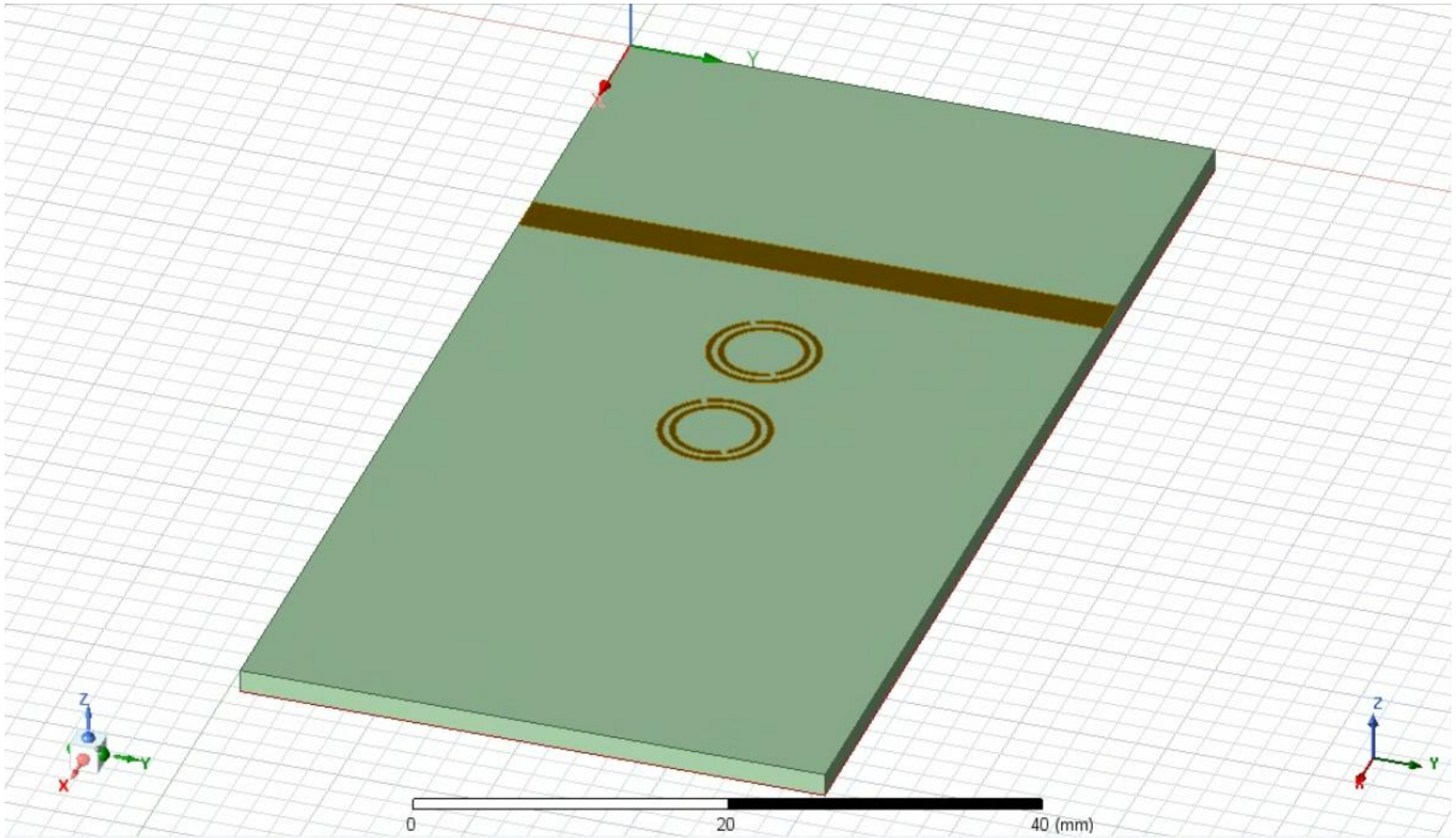


Edit model

You can edit your model with following tools

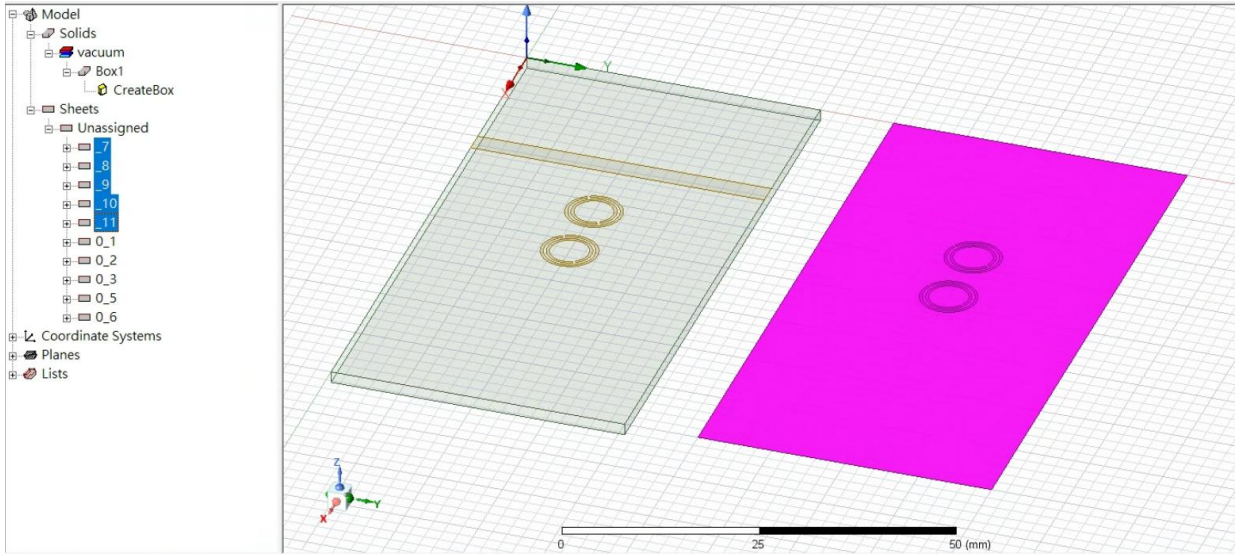


- Delete the board on DSRR side as designed(brief introduce)



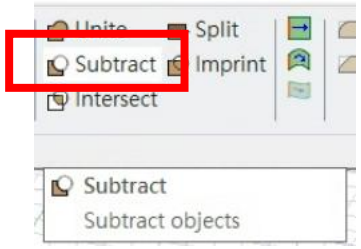
Edit model-**subtract**

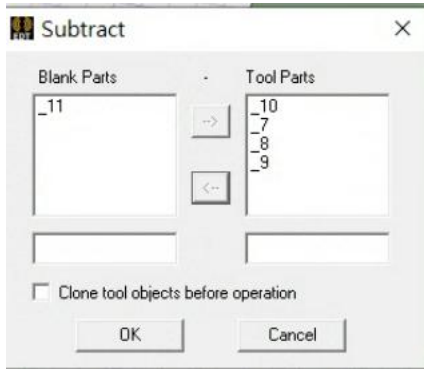
1.**Use Ctrl or shift to select** the object you want to change



You will see the color changing after select, to confirm whether that is what you want

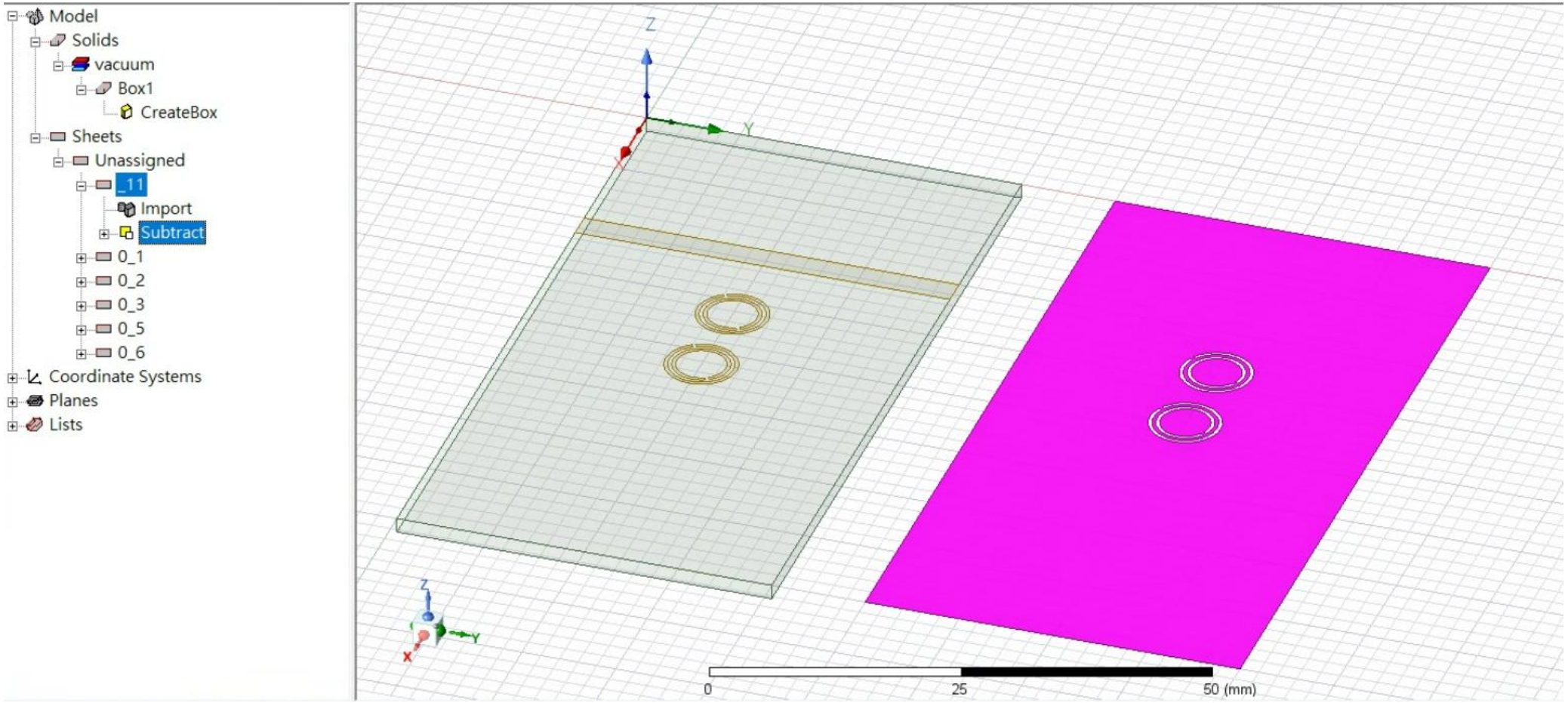
2. click **subtract icon icon-**>Blank part(board)(using left right button to edit)->ok



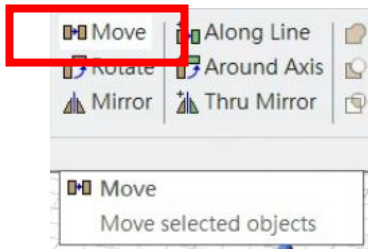


Tips : blanks part will be the remain component

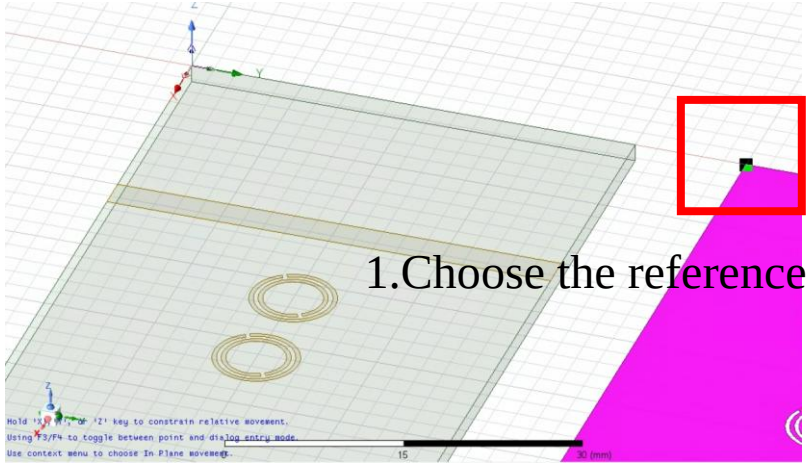
Subtract result



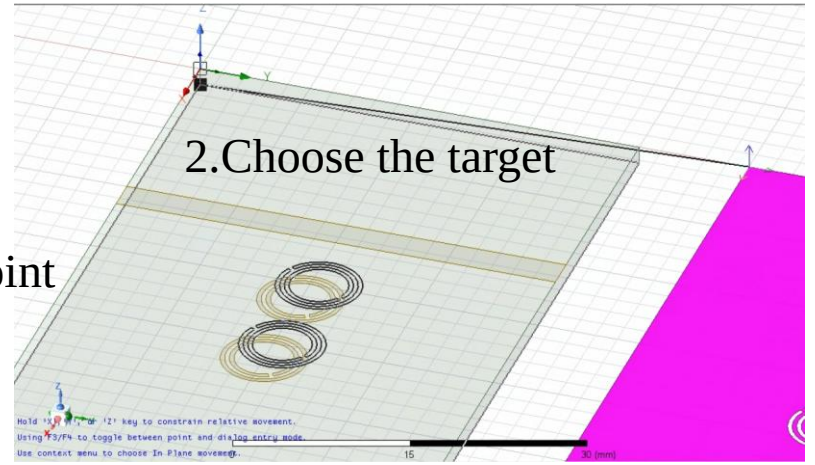
Edit model-**move**



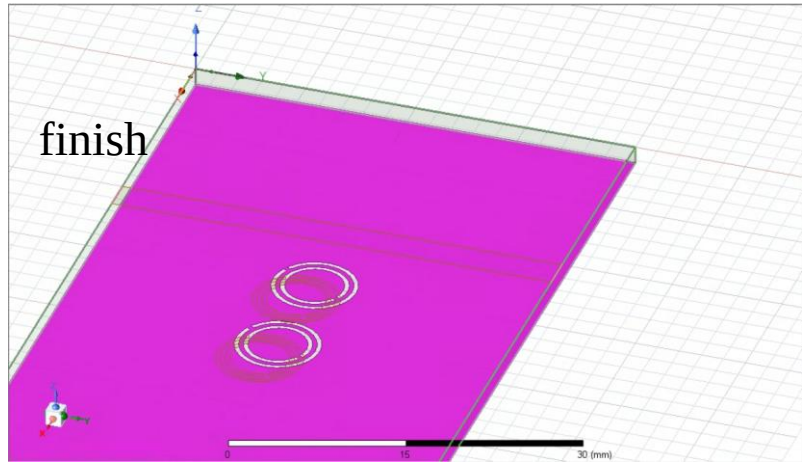
- **Select** the CSRR board one->Click **move** to place your back board



1.Choose the reference point

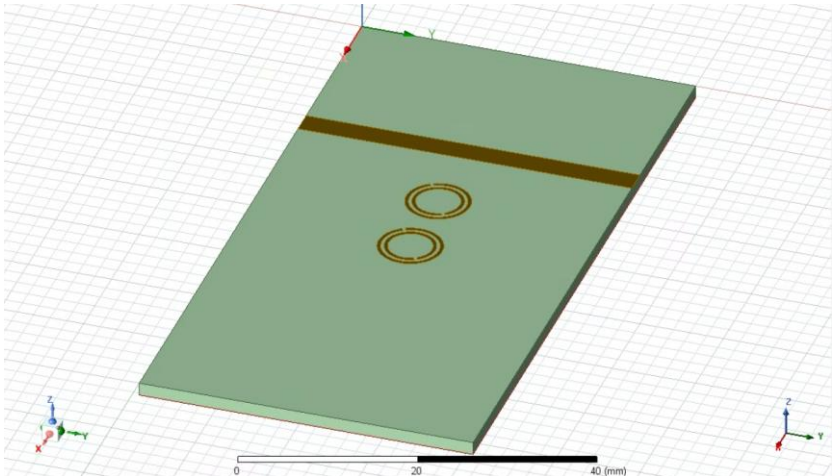


2.Choose the target



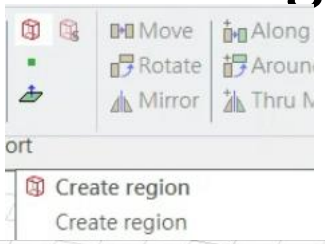
finish

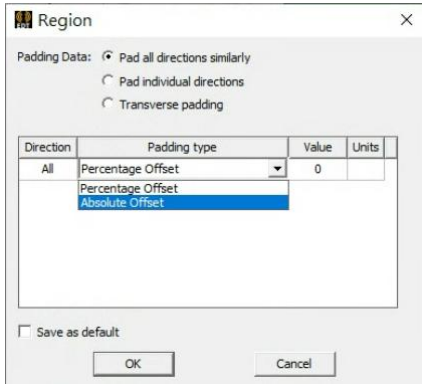
Then you will get:

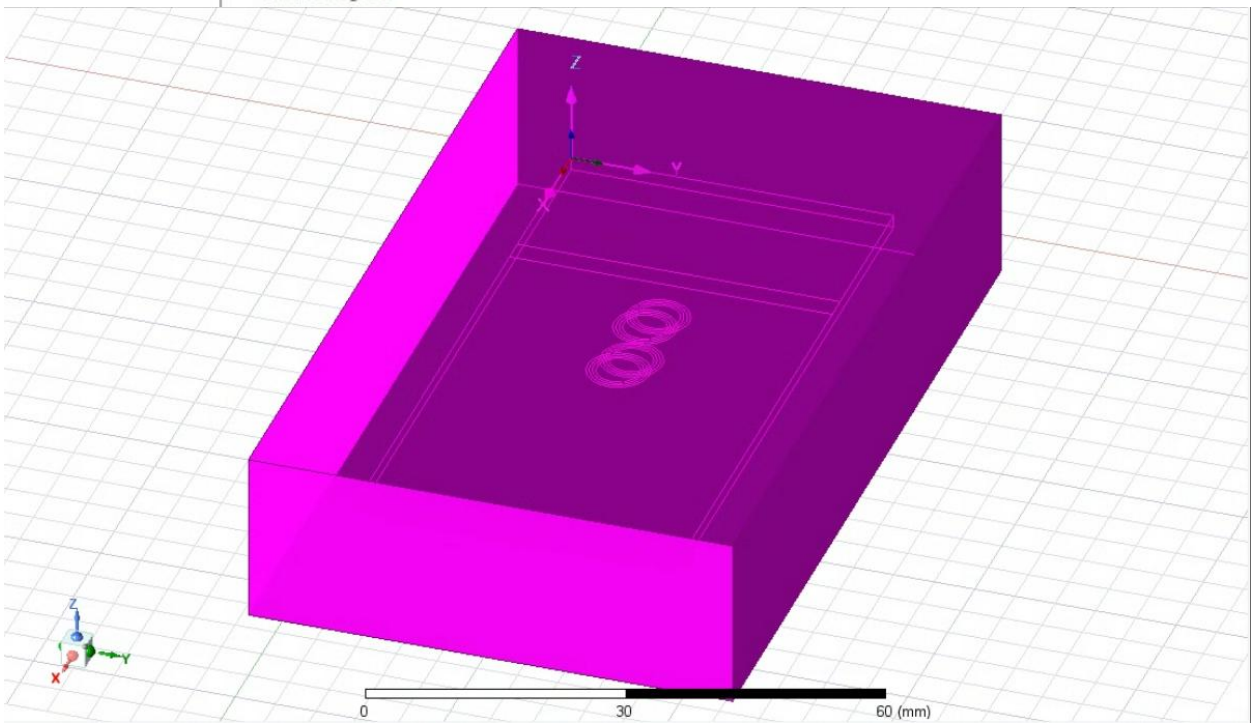


Edit model **-Create region**

- Click **create region icon-**>select **Absolute offset-**>value 10mm







This region will be used to calculate the EM result, referring where it need to calculate

Brief information about assign

- **Assign material** : define the object electromagnetic properties , only for **3D object**

Ex : PCB(in room temperature FR-4, in low temperature RO4003C),also superconductive model(thin film using thicken sheet)

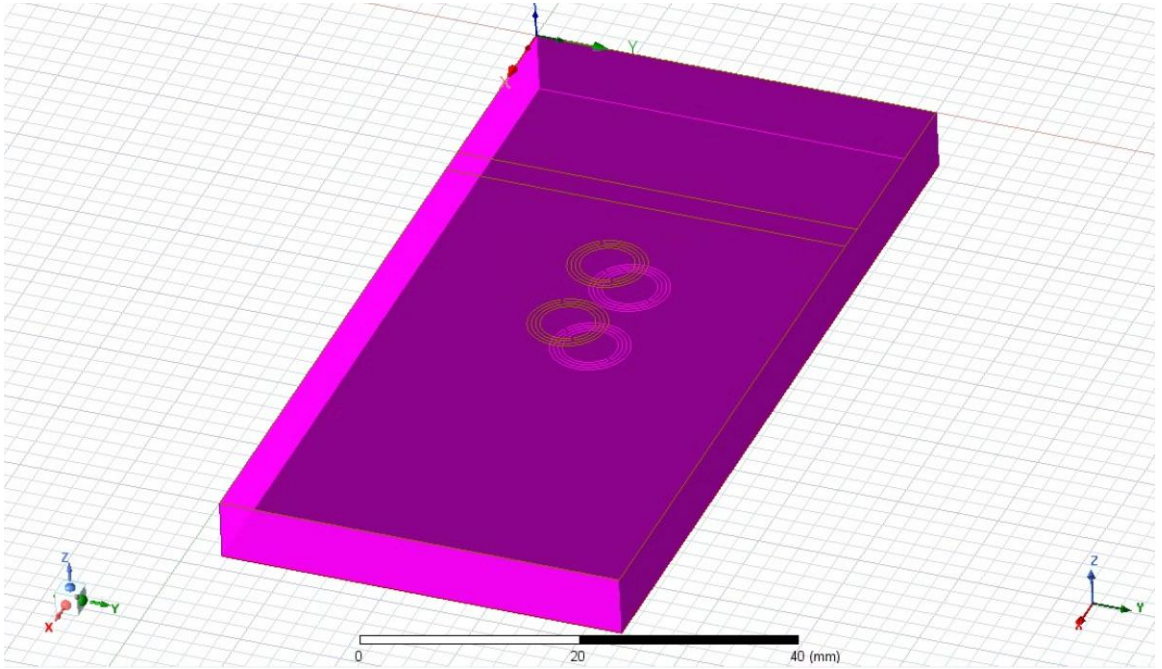
- **Assign boundary :** Boundary conditions specify the field behavior at the edges of the problem region and object interfaces.
    1. To create either an open or a closed electromagnetic model
    2. To simplify the electromagnetic or geometric complexity of the electromagnetic model

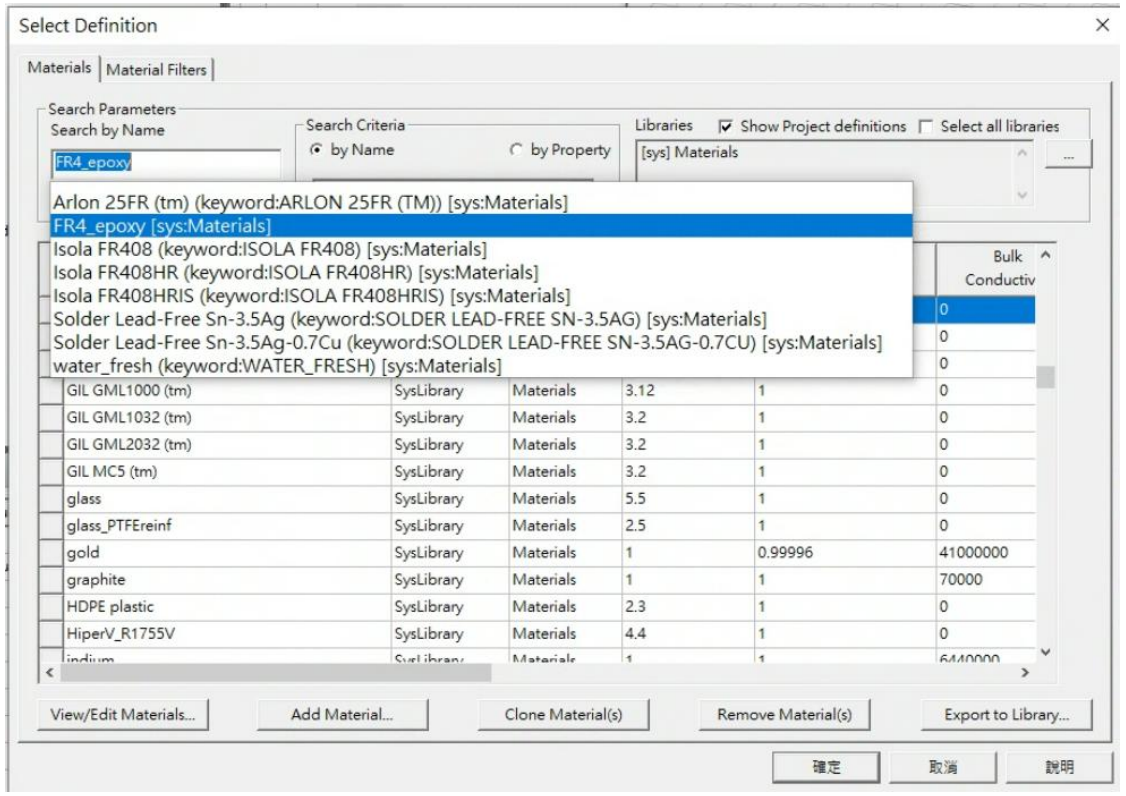
Ex : simplify model PerfectE , impedance boundary, lumped port(driven model)

Assign material

- **select the object and Right click->assign material->choose yours**

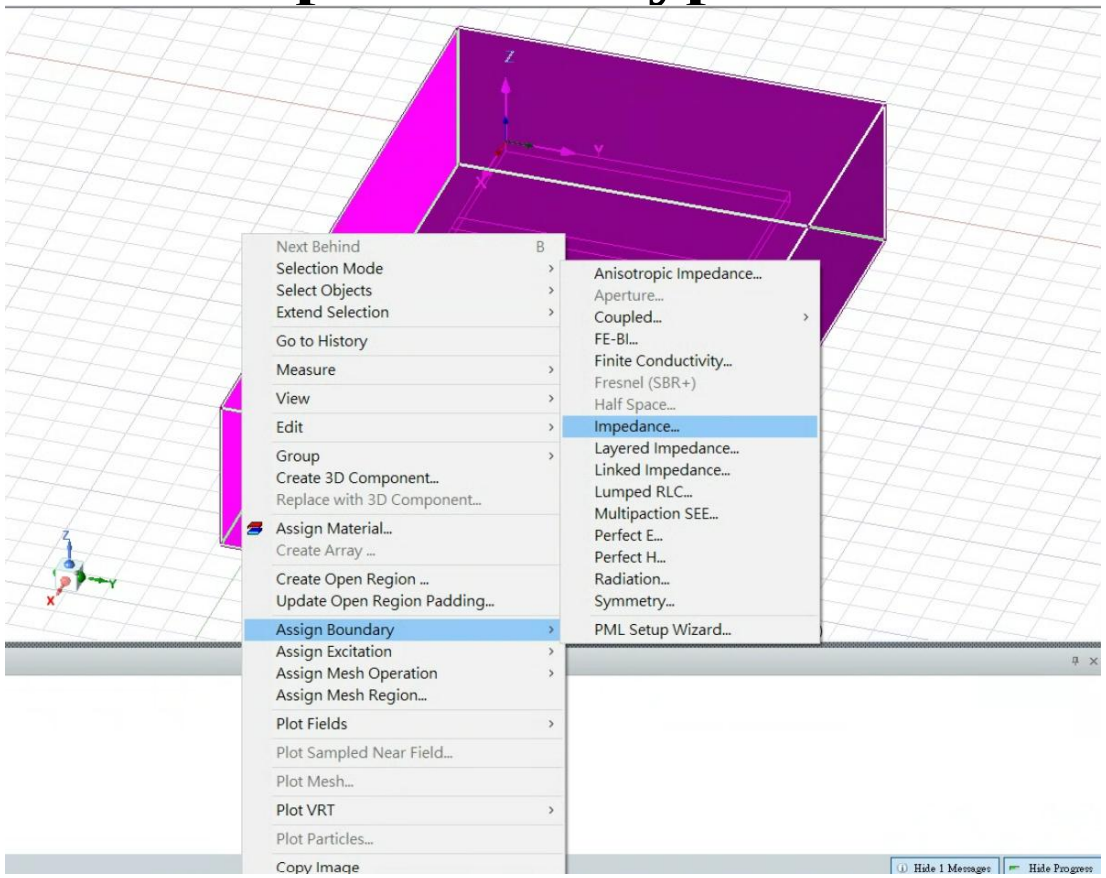
Here is PCB material so chose FR-4

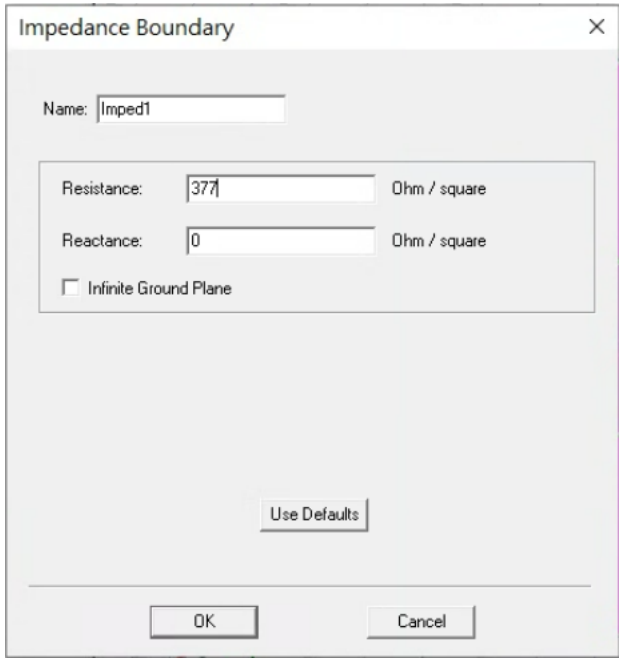




Assign boundary-free space

- **select the big region and Right click->assign boundary->impedance->type in 377 and 0**



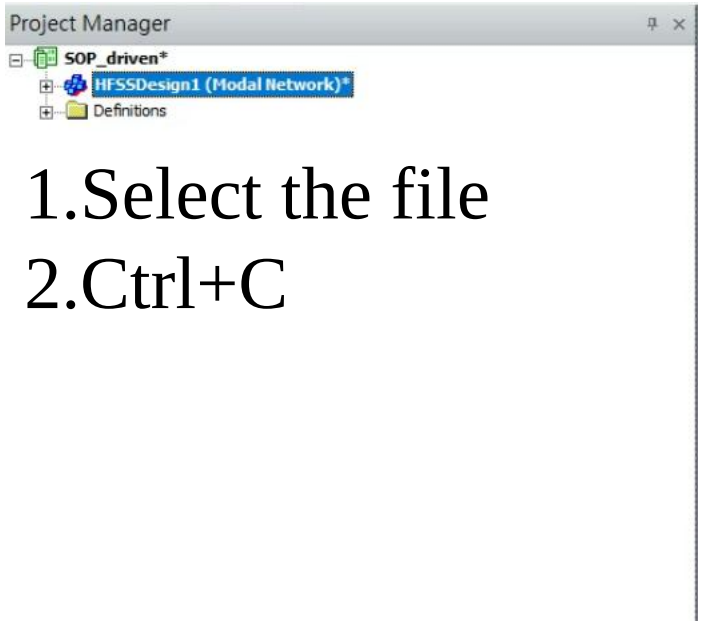


Impedance represents a resistive surface, 377 refers the **free space impedance**=120*pi

Tips : after assigning you can right click->view->hide selection

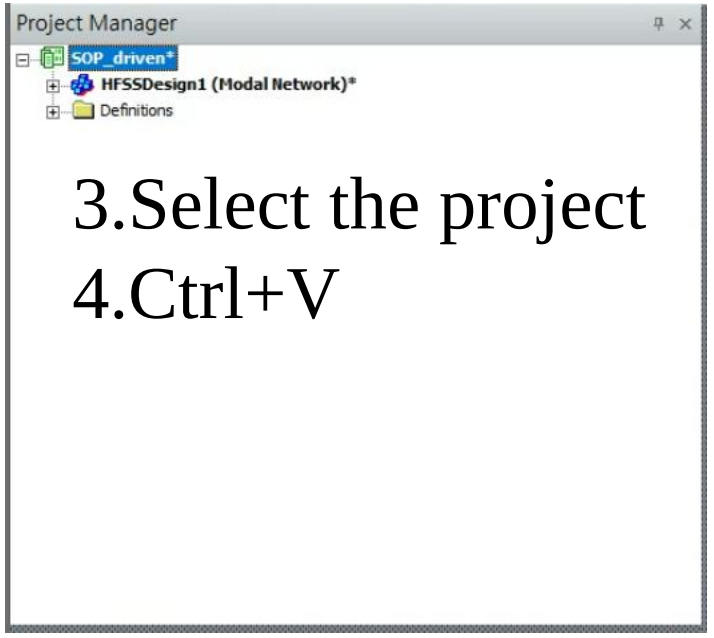
Caution!!!

- Here we will separate into two set up methods of driven model and eigenmode , but don’t worry the model you had set up, because the setting before is alike. What you need to do is copy and paste and keep going



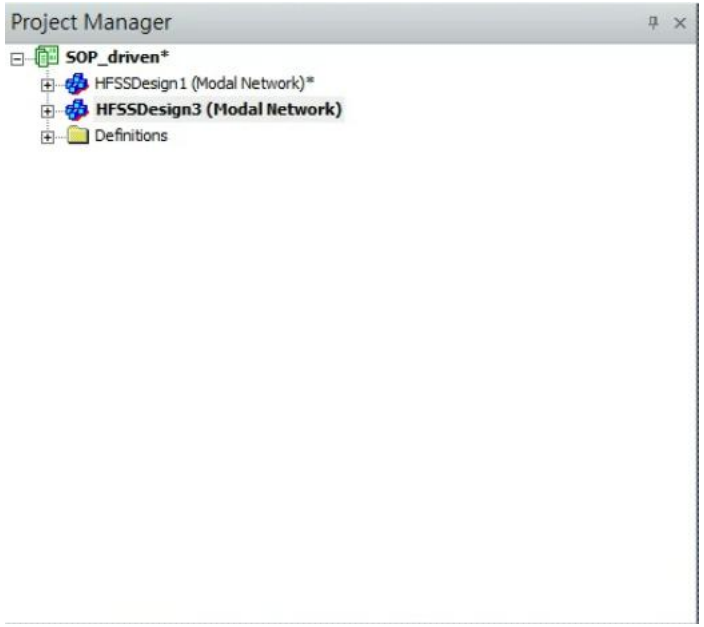
1.Select the file

2.Ctrl+C



3.Select the project

4.Ctrl+V



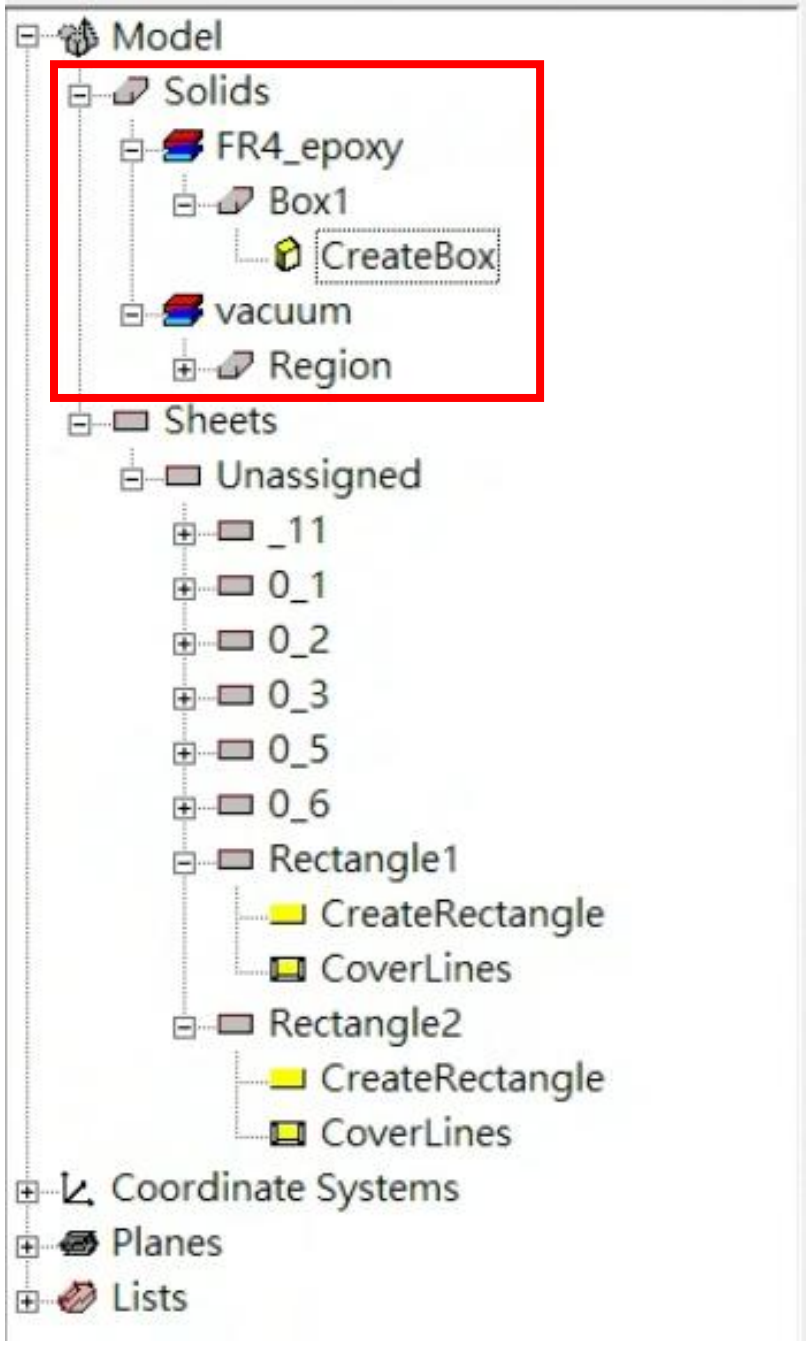
Driven model

Assign material

- Use the tips before:

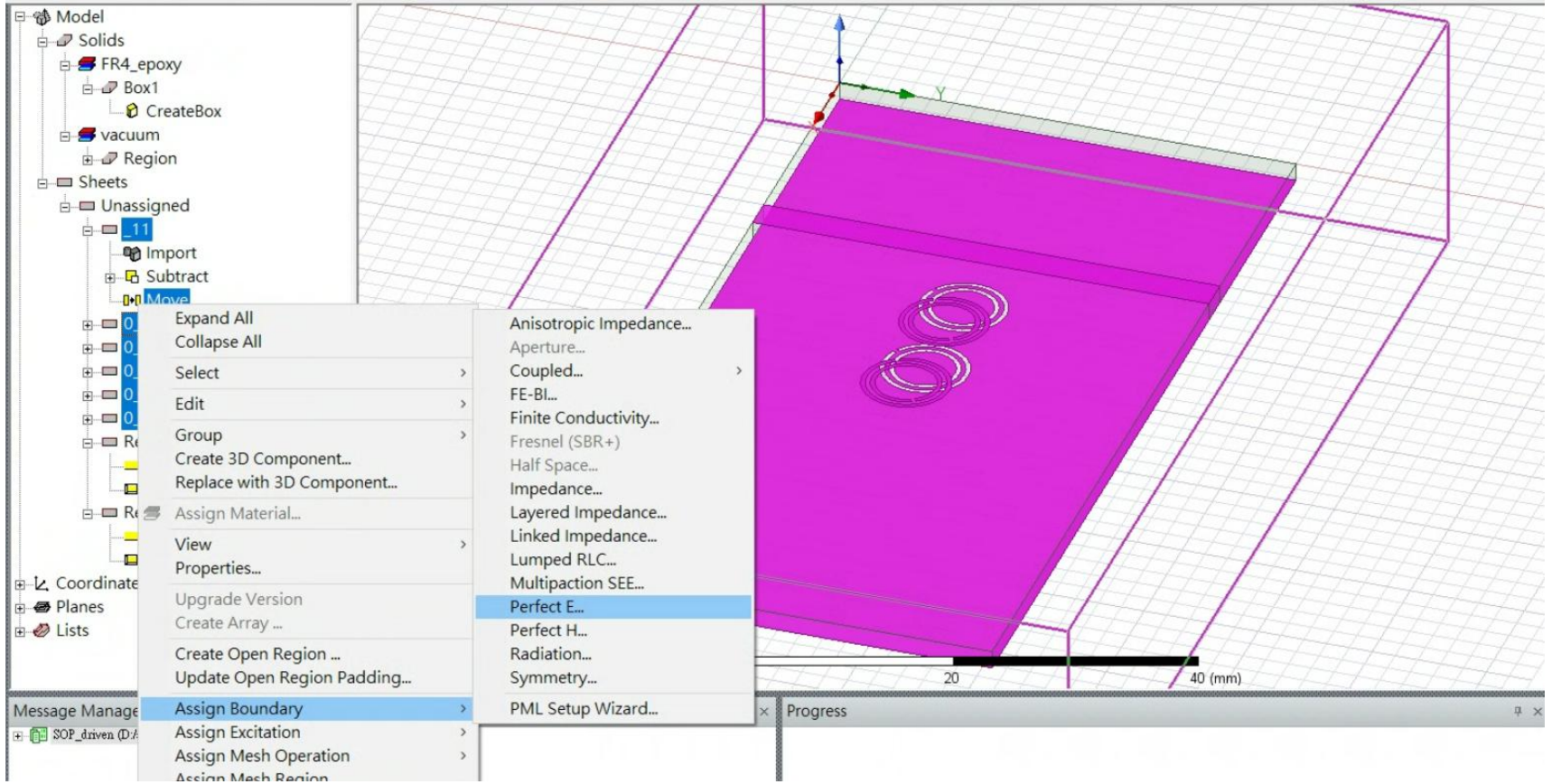
Assign the PCB as **FR4**

Assign the free space as **vacuum**



Assign boundary

- Select the DSRR , transmission line and **right click-> assign boundary->PerfectE**



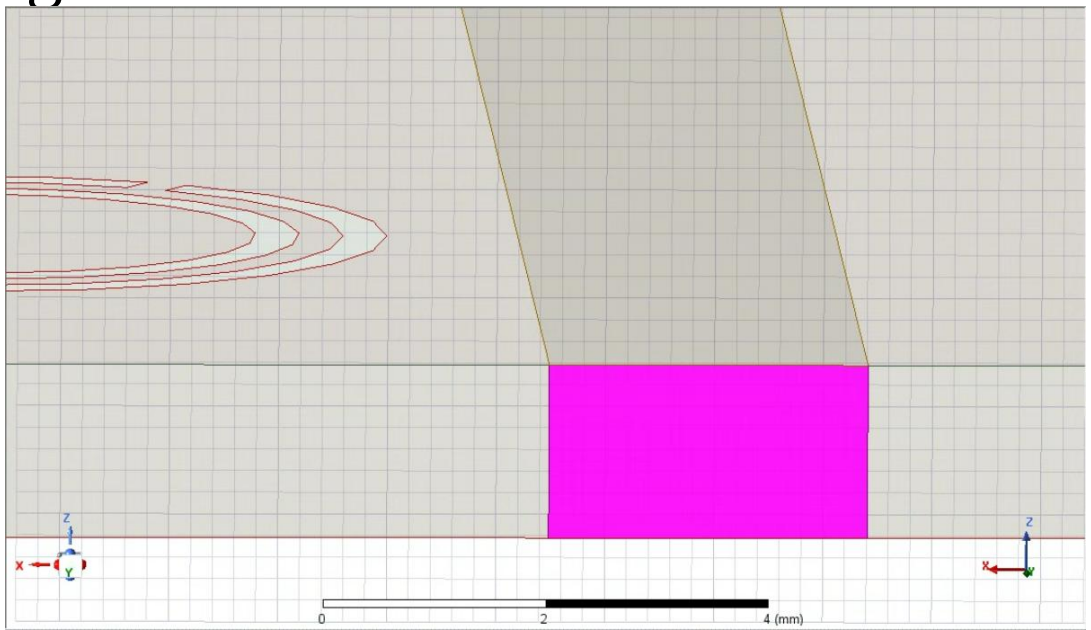
Assign lumped port

- Lumped ports support single mode excitations when S-parameters have to be extracted at internal locations of a model. It can also be used to represent a terminal of a passive component to be subsequently optimized in a circuit simulator. Simple to say, it represent the port for the transmission to go into the resonator

- Create a surface at the left and right surface like this

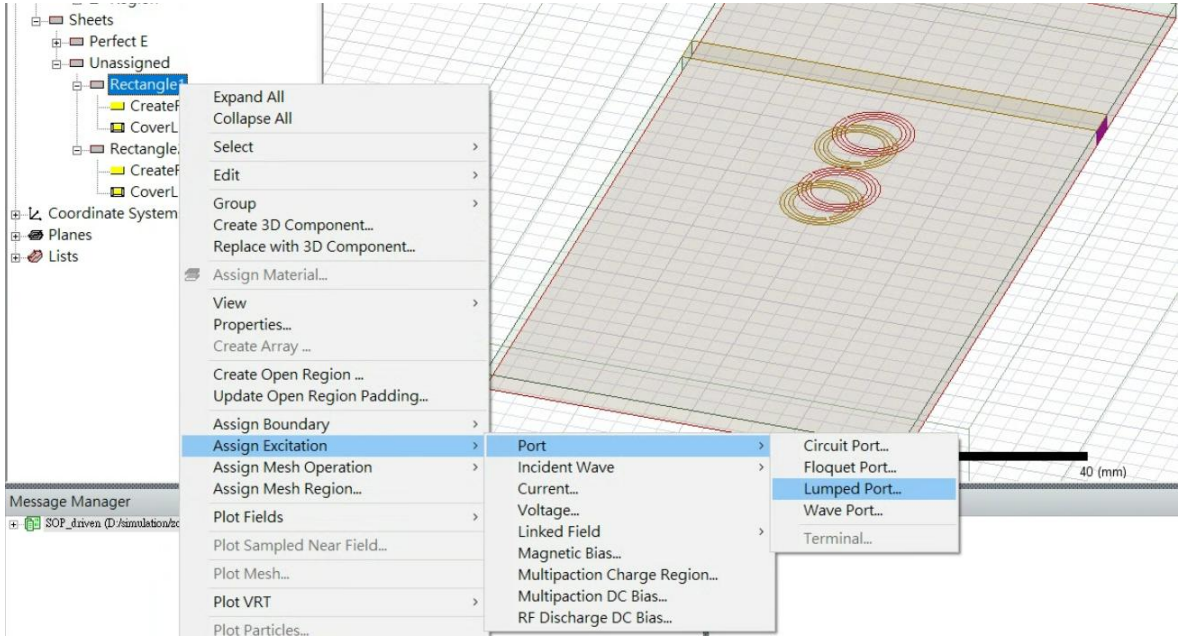


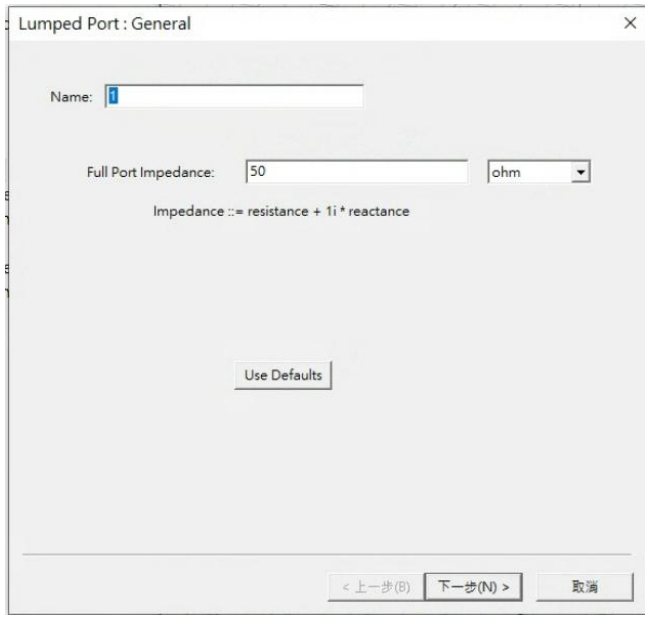
Tips : you can change your to xz-plane to create



Assign lumped port

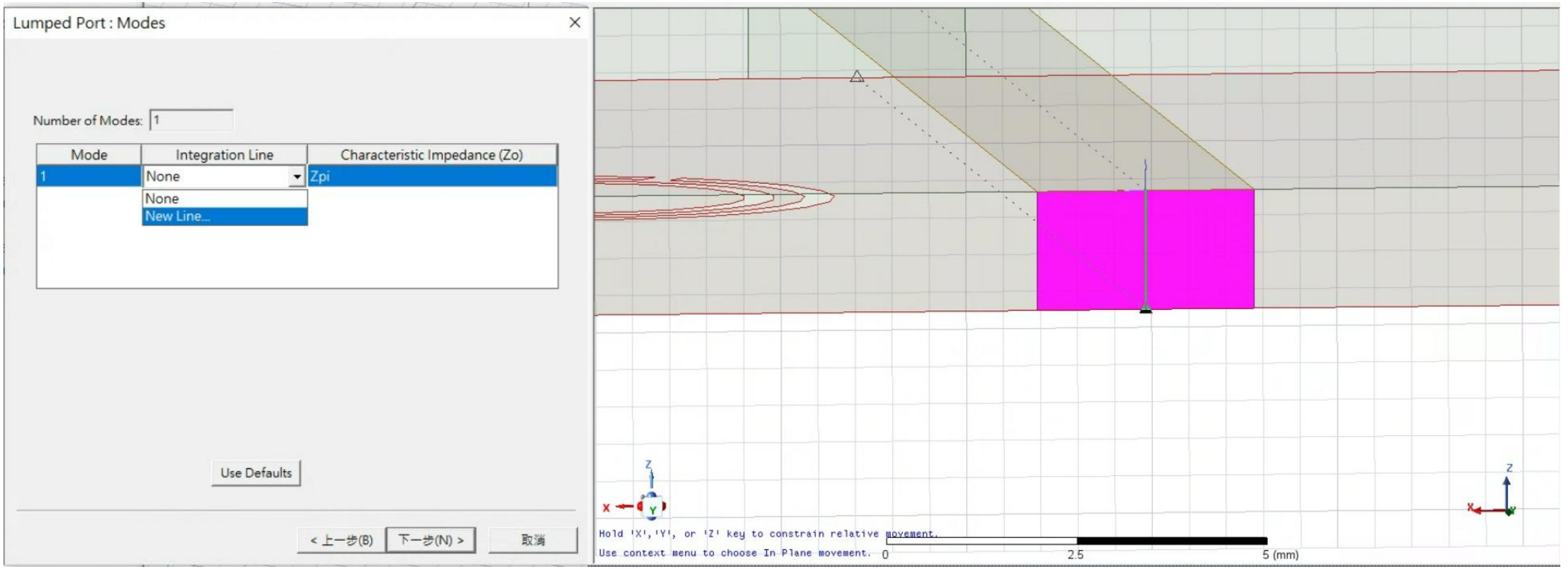
- Right click the rectangle->assign excitation->port->lumped port->next step





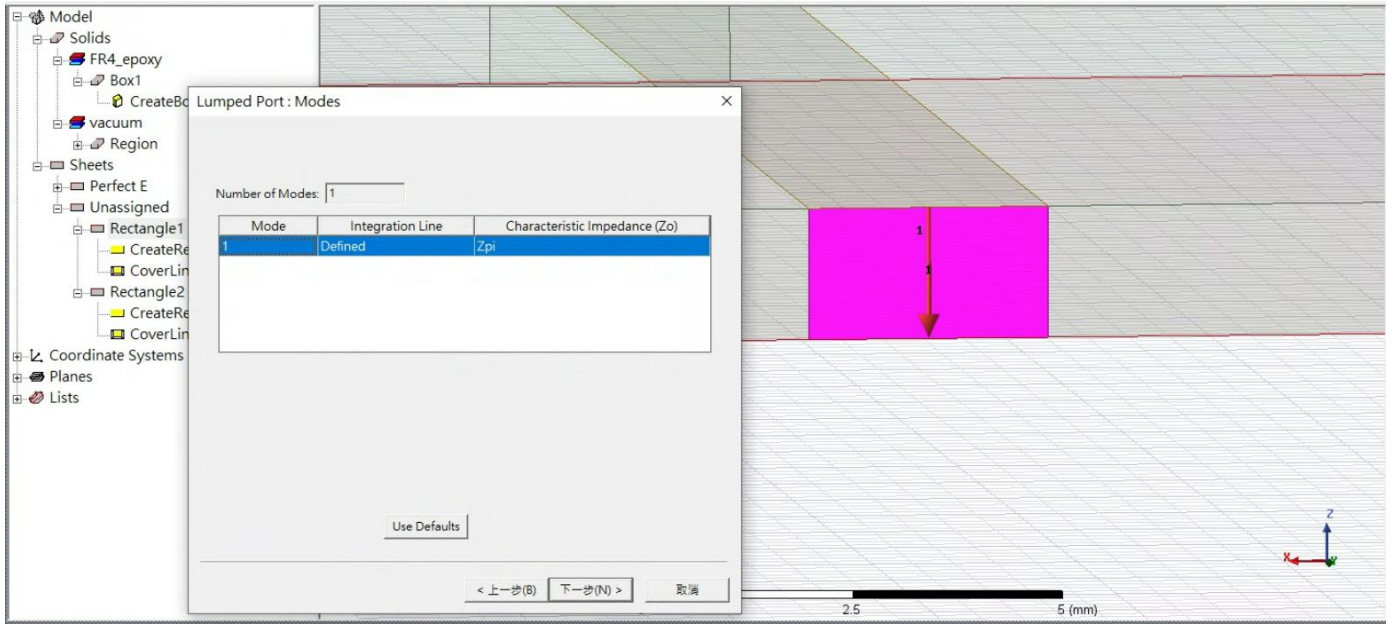
Assign lumped port

- Click new line->direction from top to down

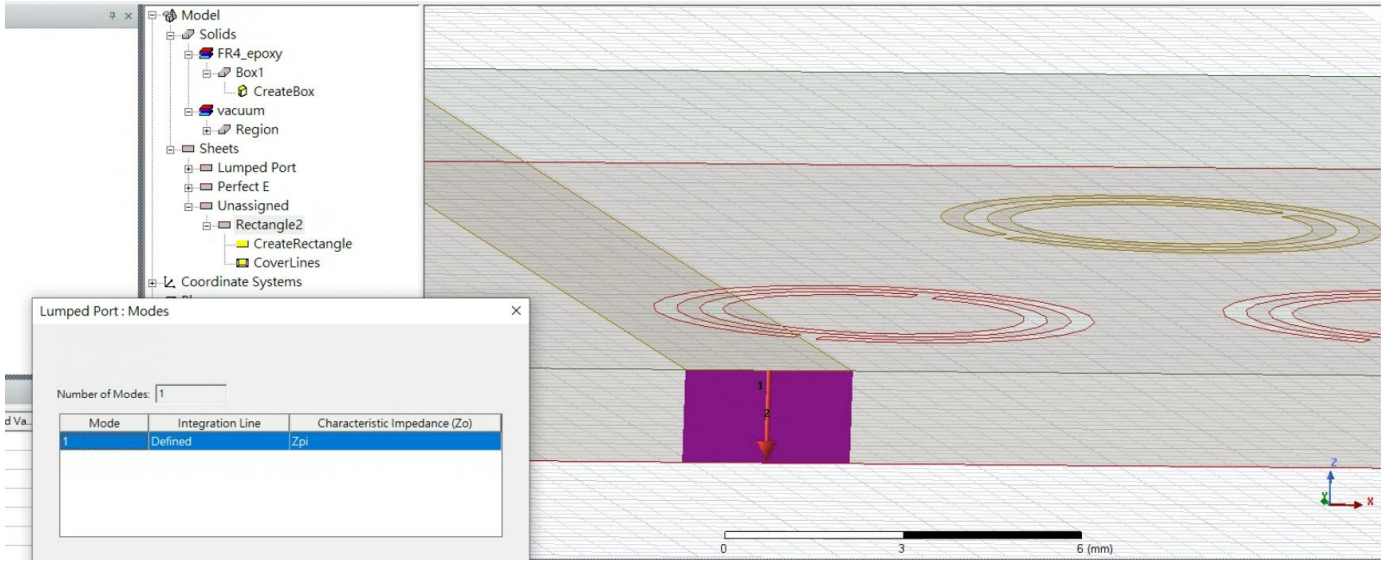


Assign lumped port

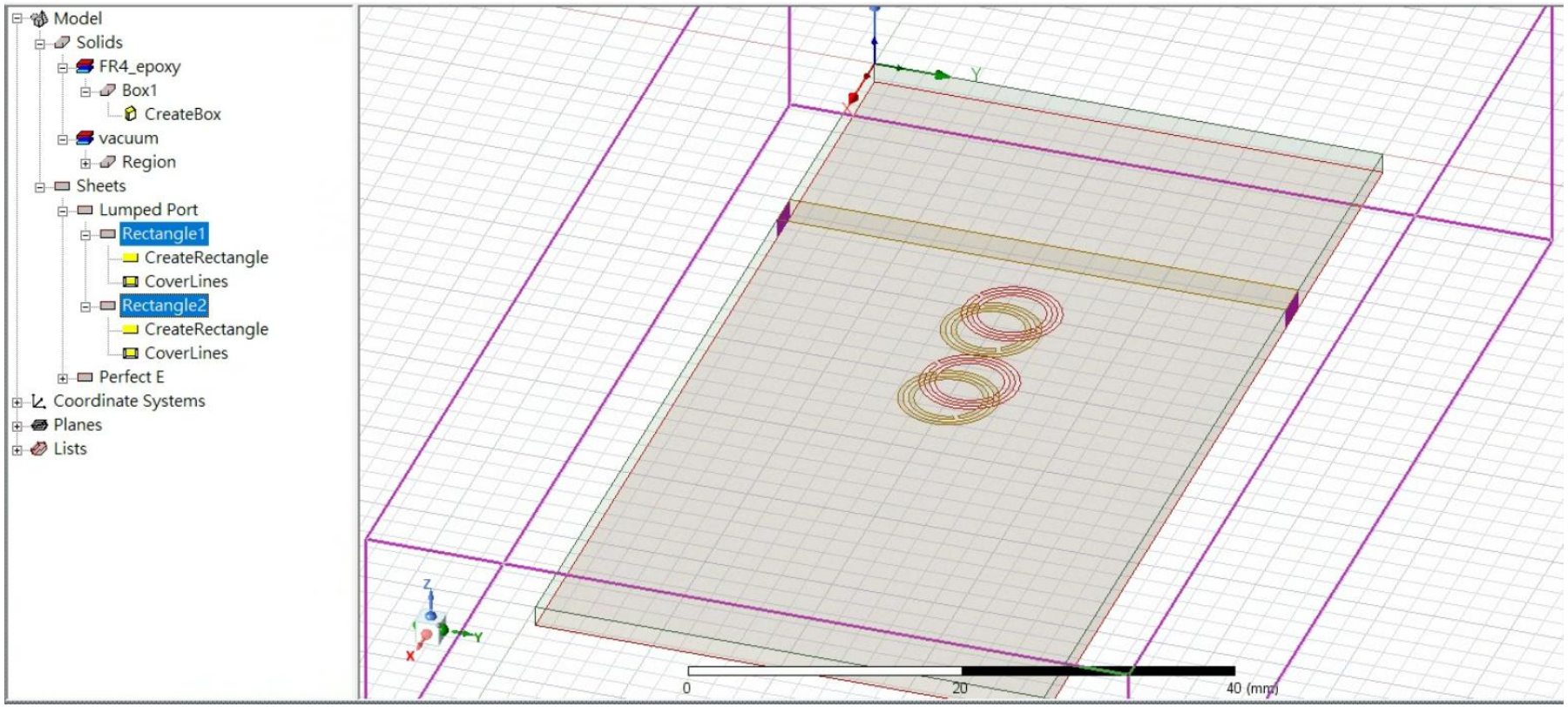
- Click 下一步 ->finish

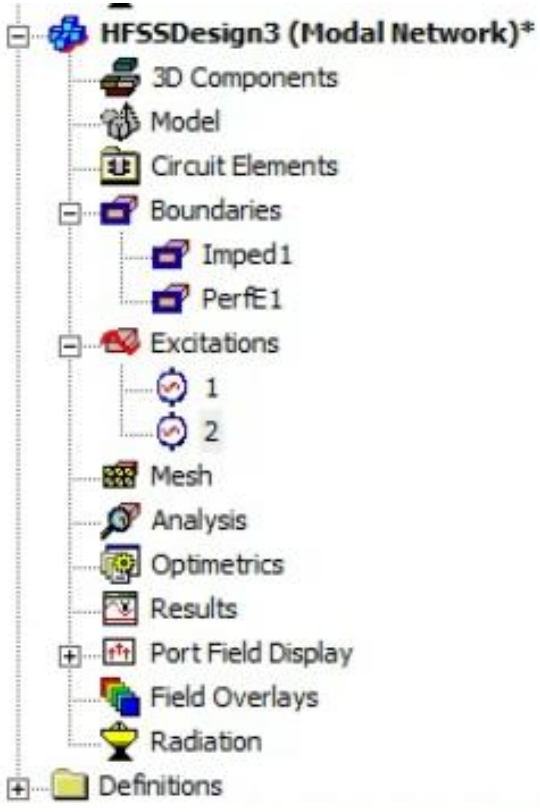


- Finish the left one too!



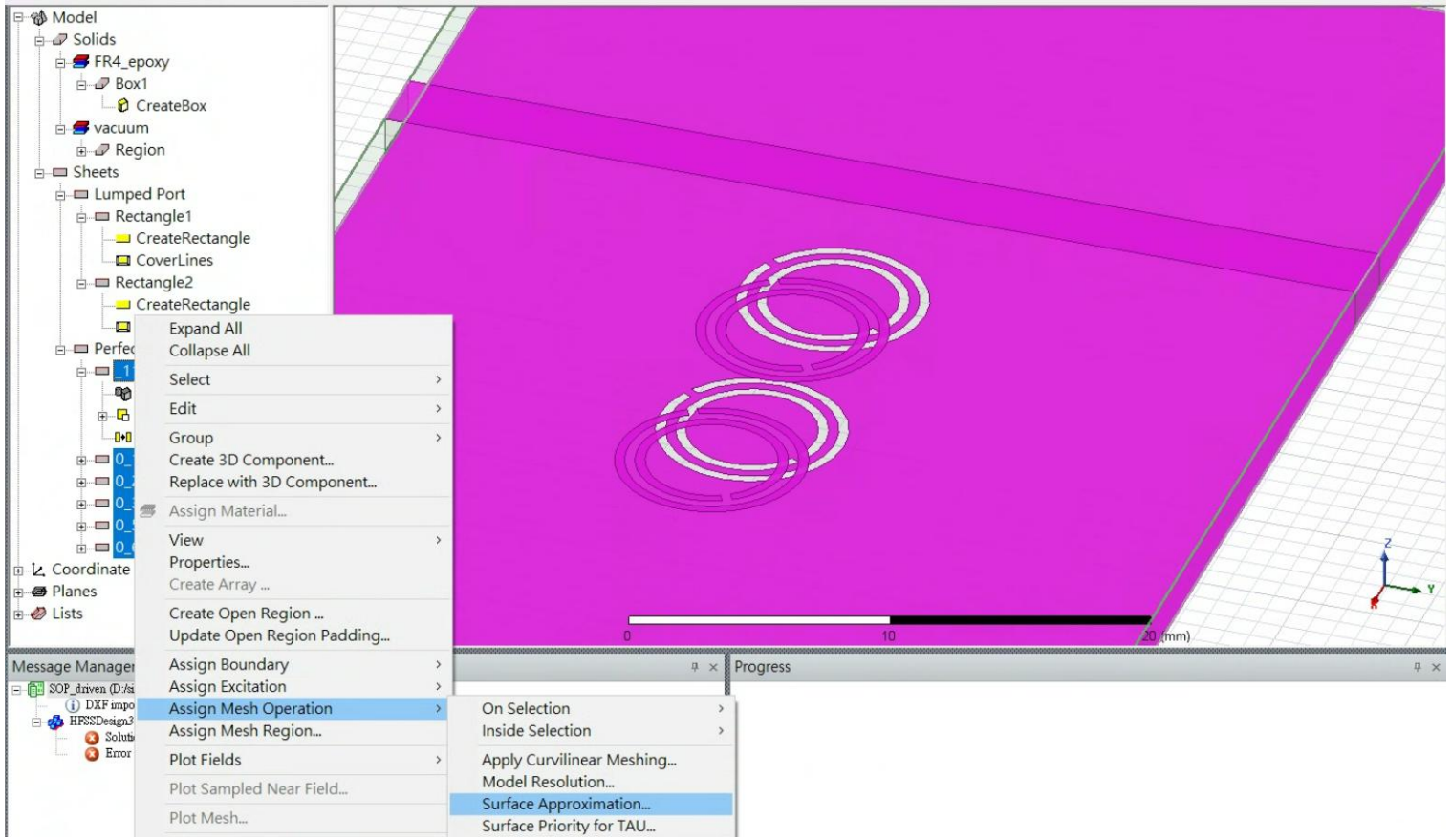
Assign lumped port- confirm



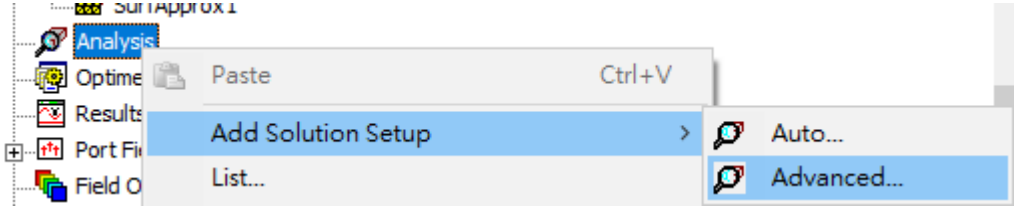


Assign mesh operation

->choose 6(if data needs to be more accurate)



Set up analysis



General

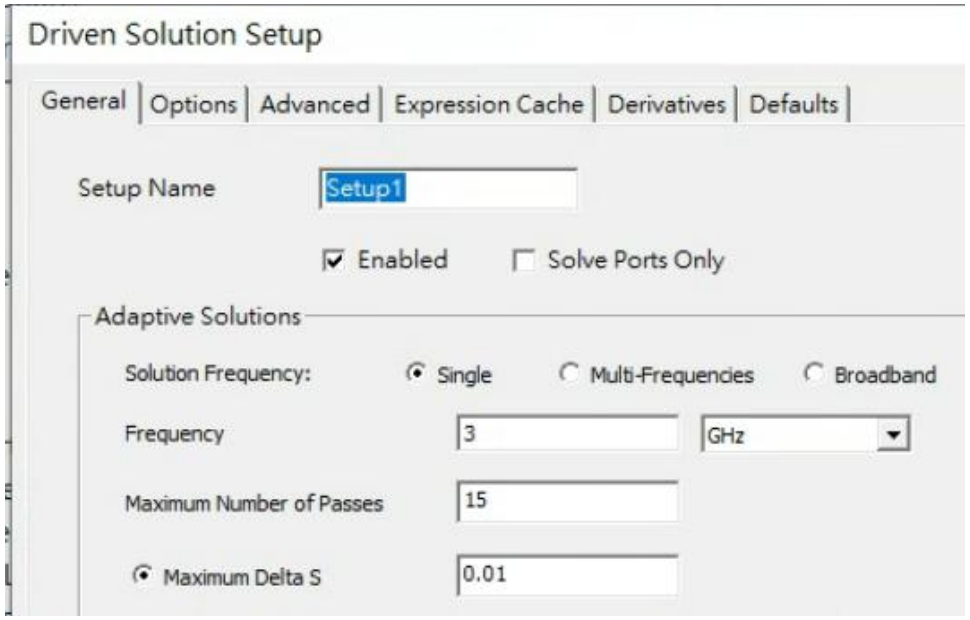
Frequency=3GHz(intermediate freq)

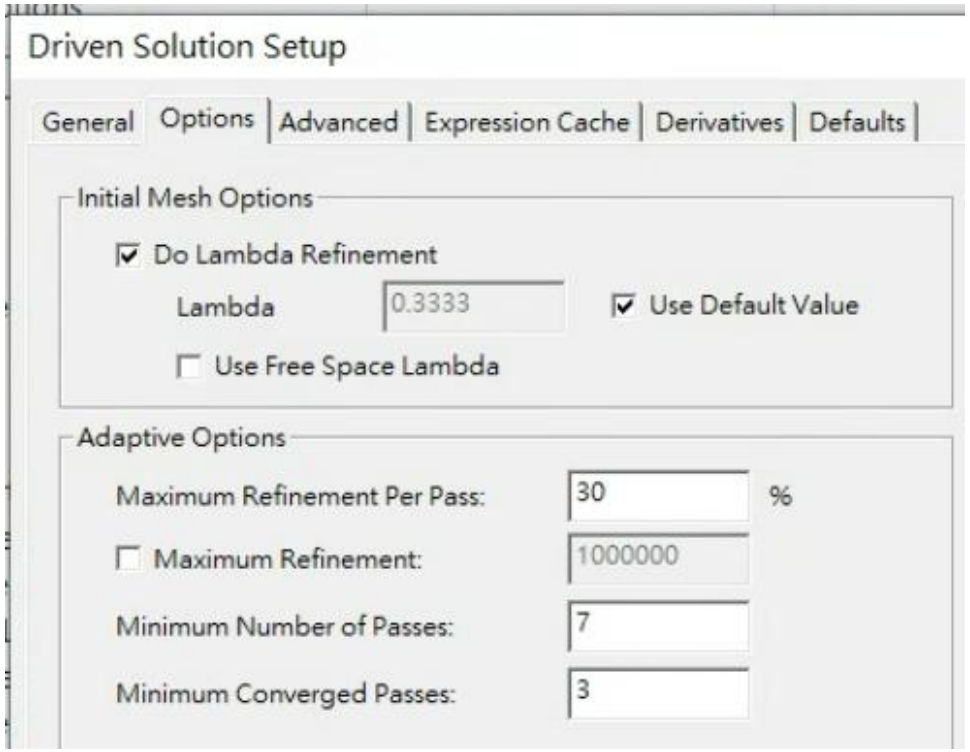
Maximum Number of passes:15

Maximum Delta S:0.01%(error percent)

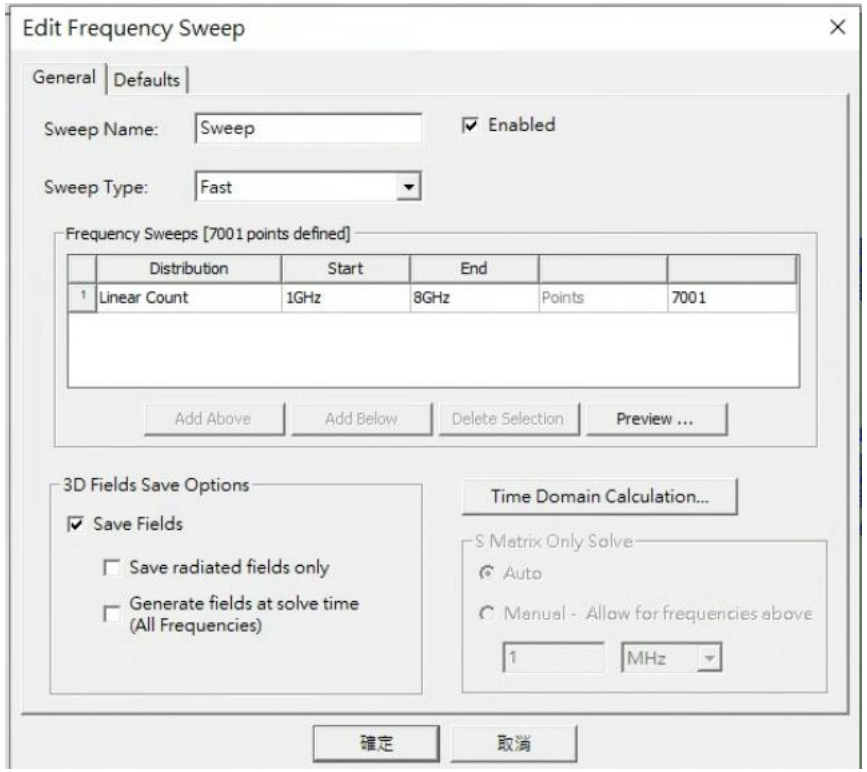
Minimum Number of Passes:7

Minimum Converged Passes:3

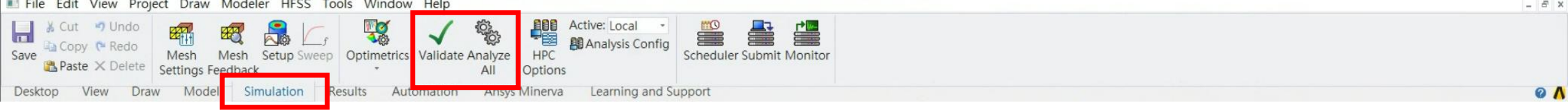




Sweep type

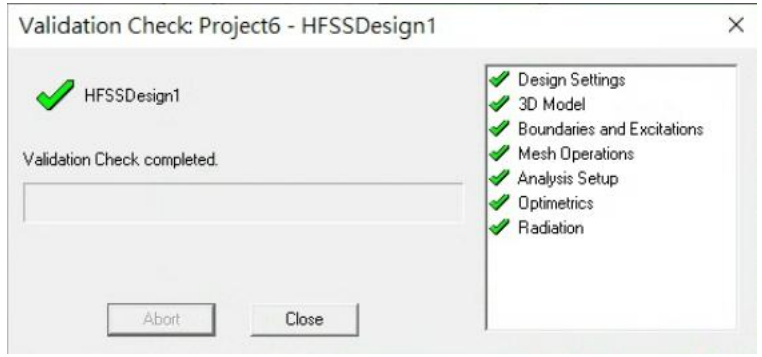


- Sweep type :fast
- point:7001pt(1MHz/per point)
- Sweeping big range:1-8GHz

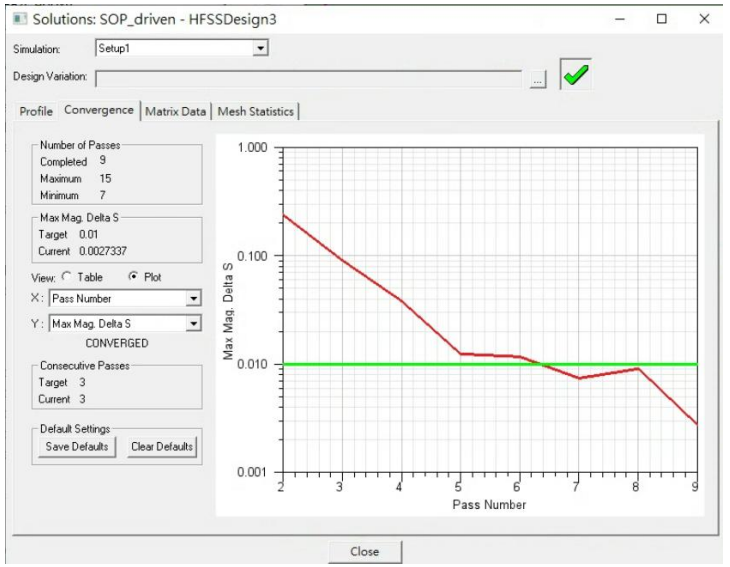


Run simulation

- Click simulation->Validate, be sure all green check
- Click Analyze All

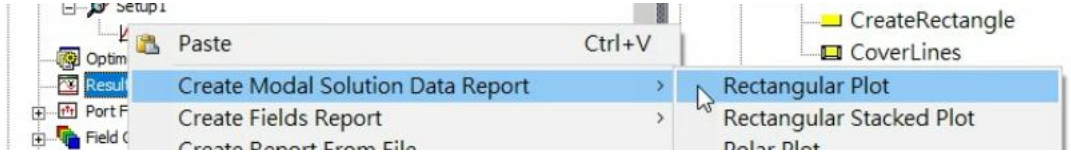


Confirm the result

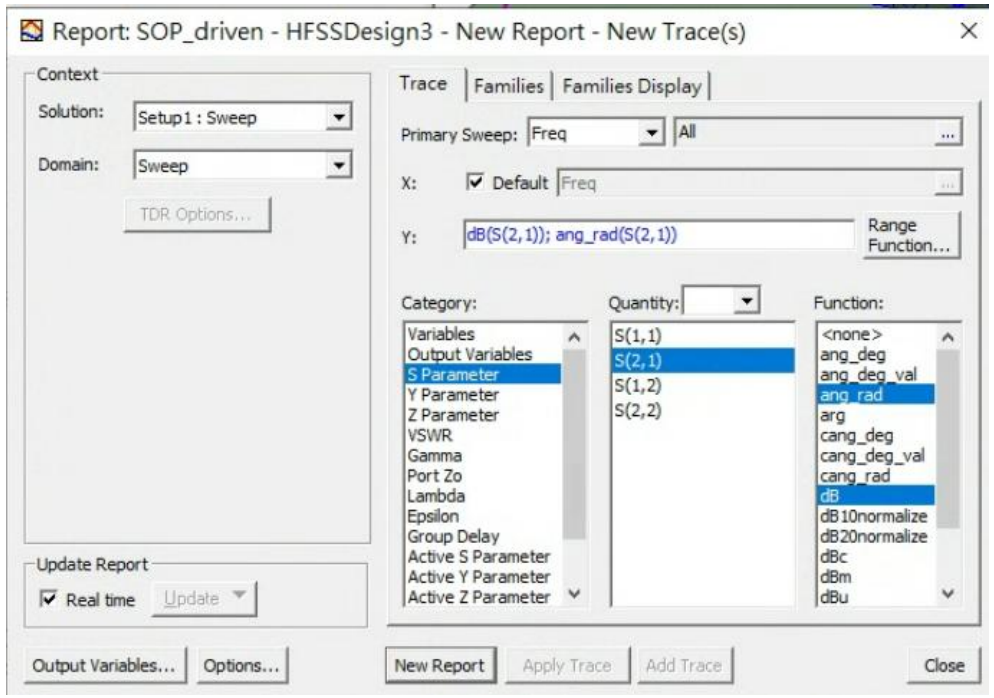


- Right click result->solution data->convergence
- Plot S(2,1)

Right click result->create modal solution data report->rectangle plot

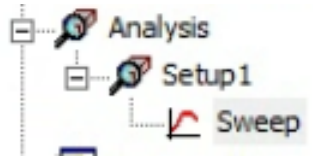


- Click S(2,1) , db , phase_rad ->new report



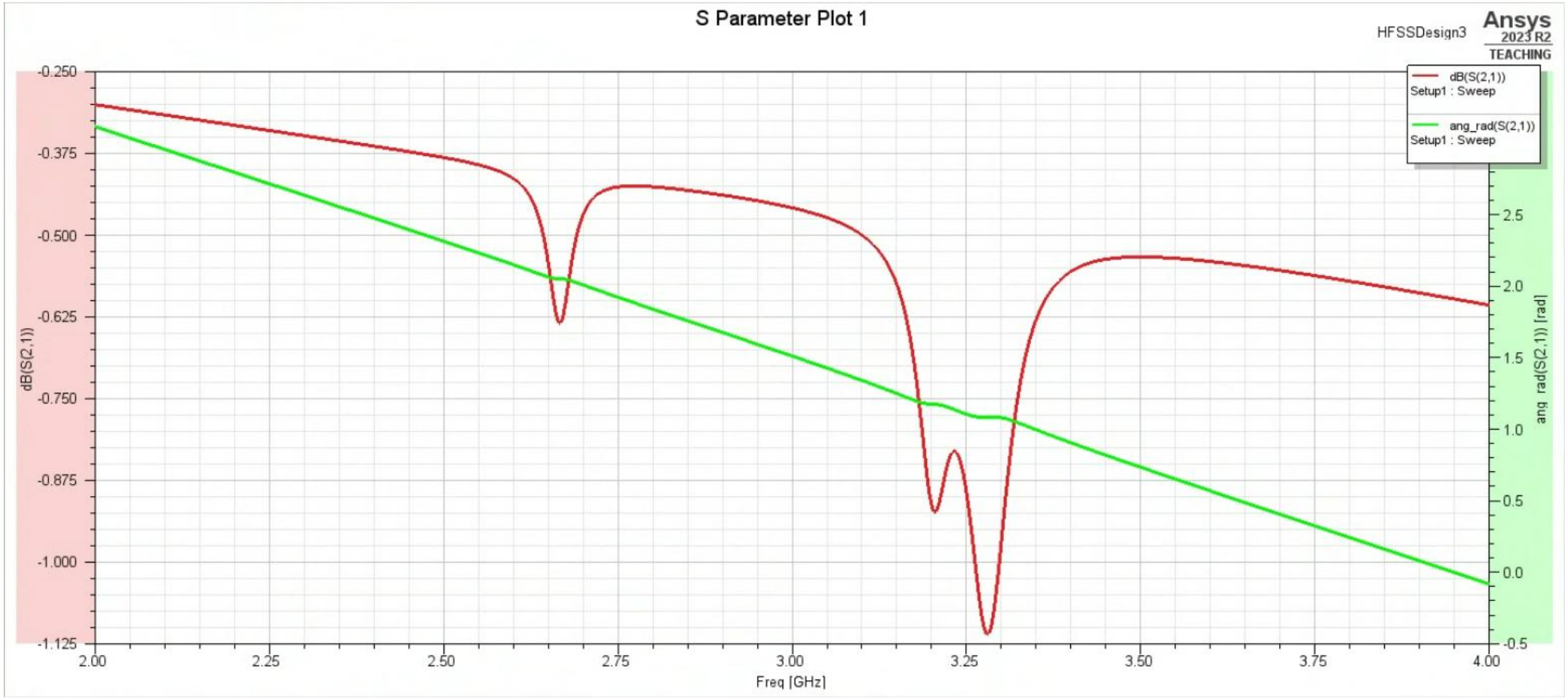
S(2,1) big range result

Here will be the one I want , change the sweeping range to small



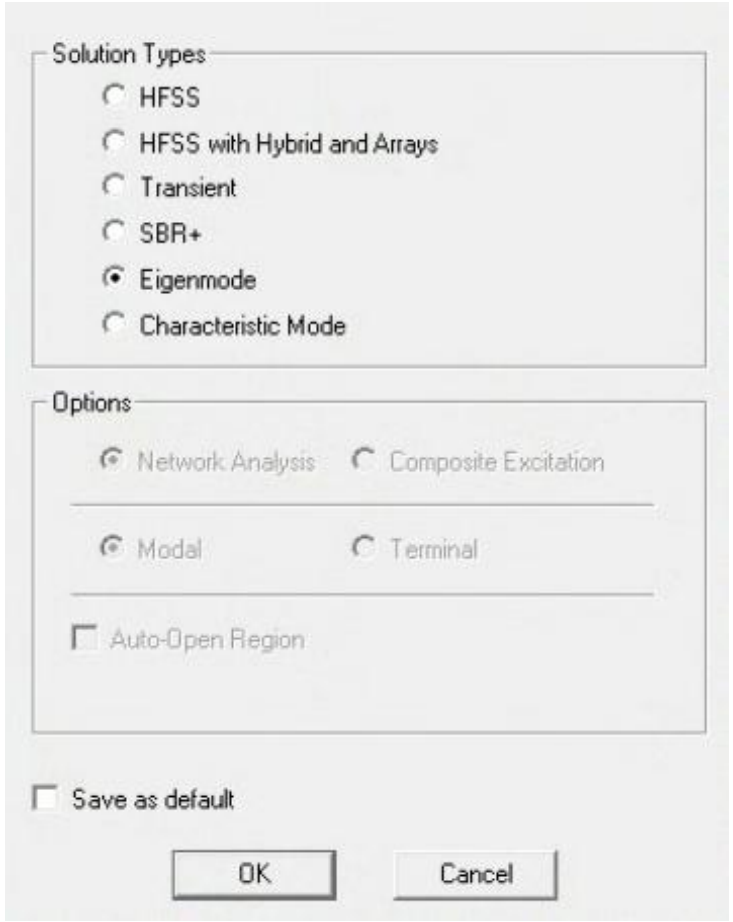


S(2,1) small range result 2-4GHz



Eigenmode

Solver ensure



Assign material

- Use the tips before:

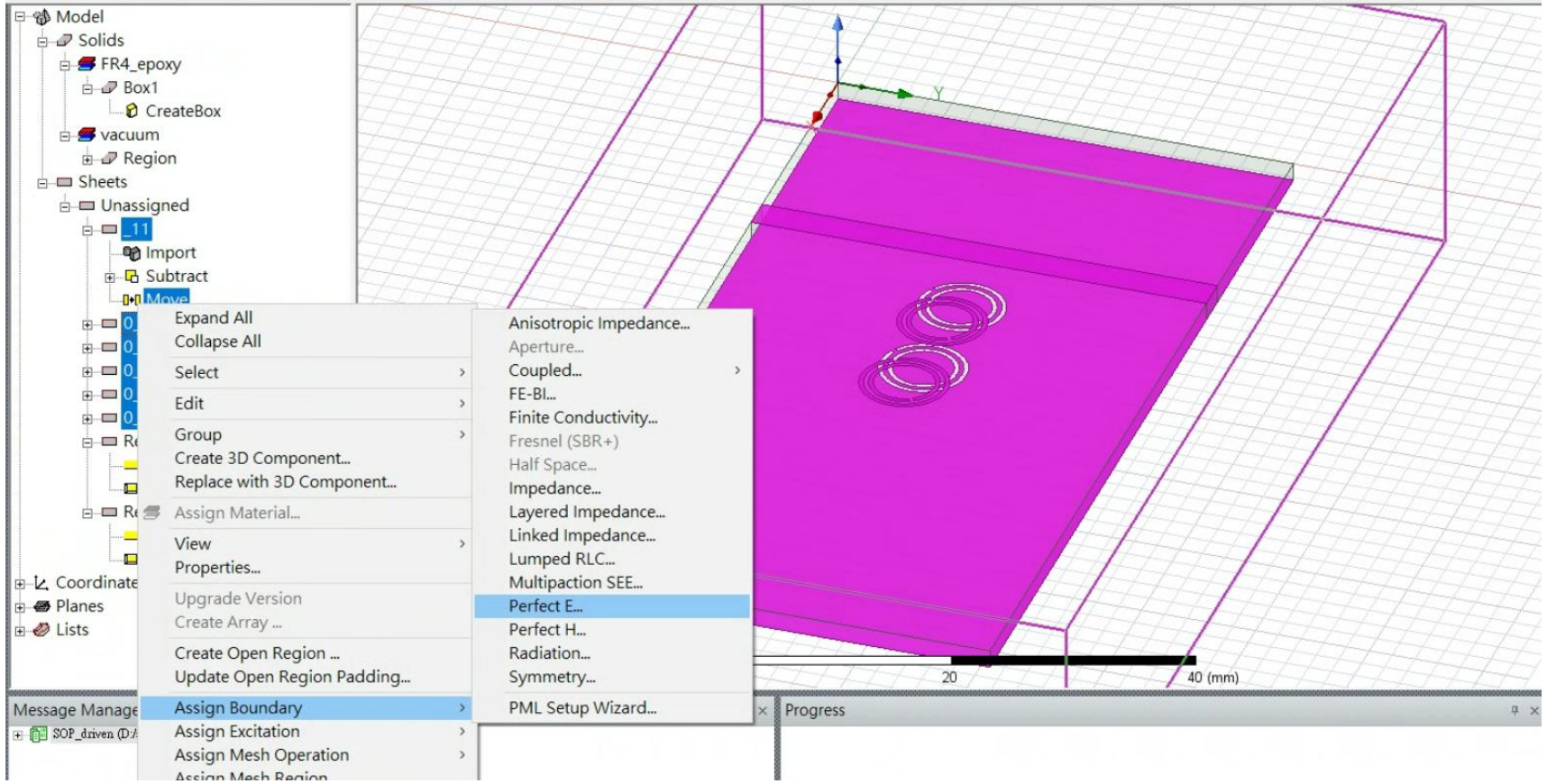
Assign the PCB as **FR4**

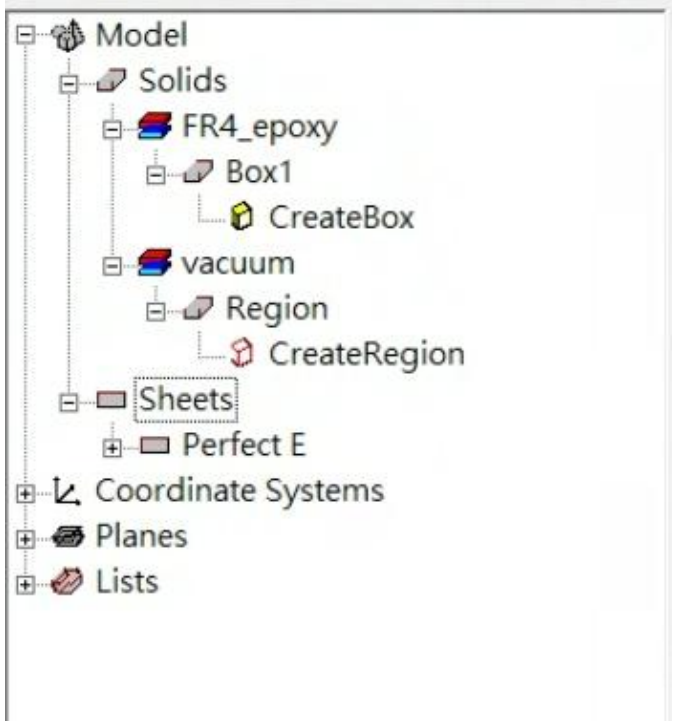
Assign the free space as **vacuum**



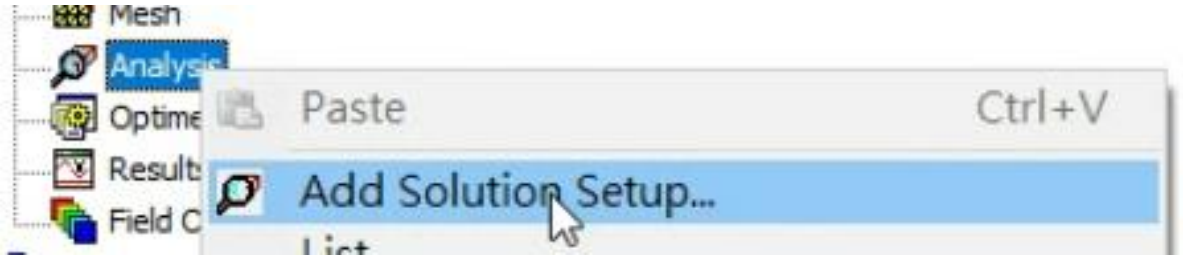
Assign boundary

- Select the DSRR , transmission line and **right click-> assign boundary->PerfectE**





Set up analysis



General

Frequency=3GHz(intermediate freq)

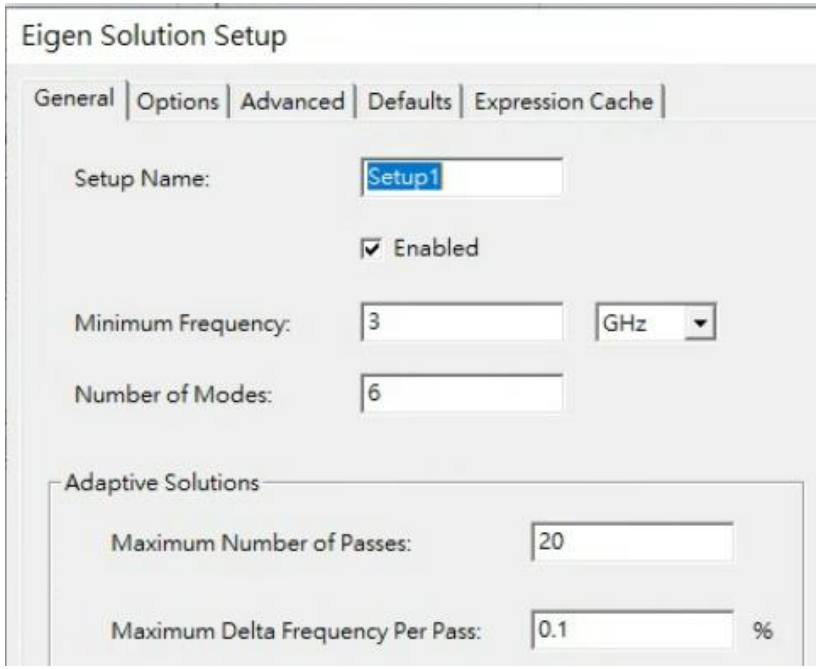
Numbers of modes:6

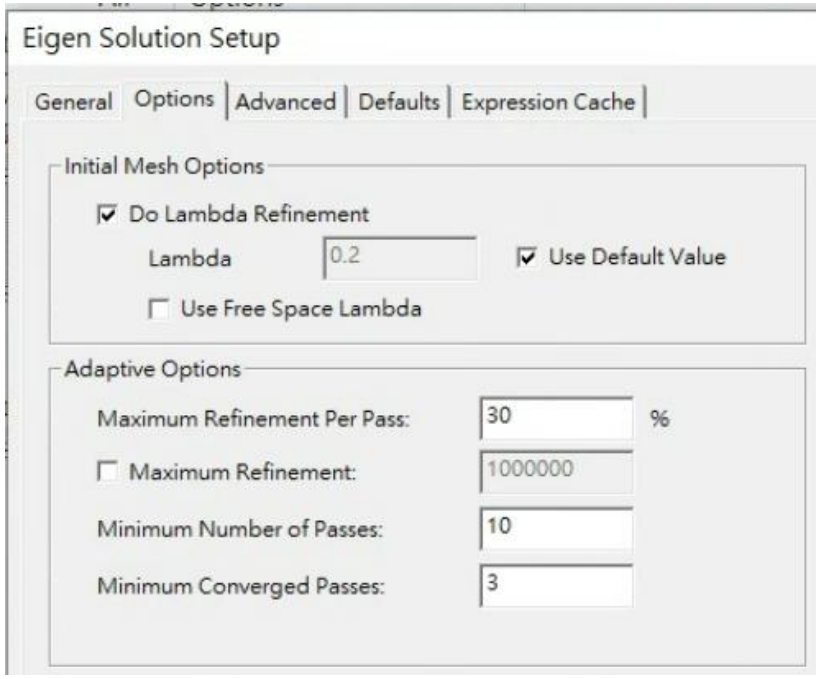
Maximum Number of passes:20

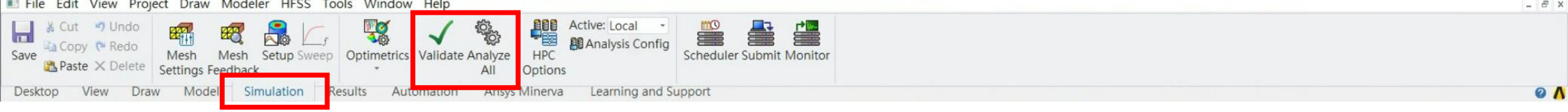
Maximum Delta S:1%(error percent)

Minimum Number of Passes:10

Minimum Converged Passes:3

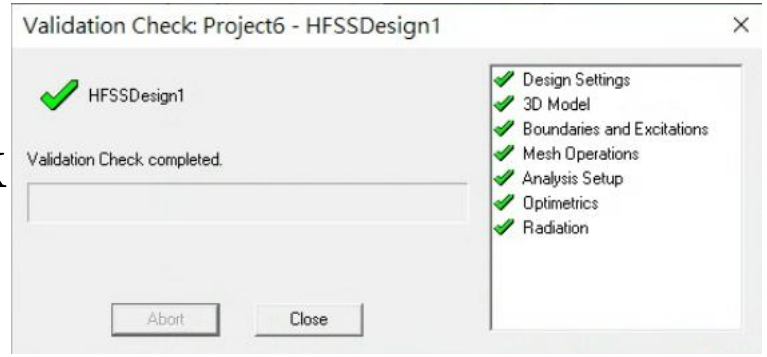






Run simulation

- Click simulation->Validate, be sure all green check
- Click Analyze All

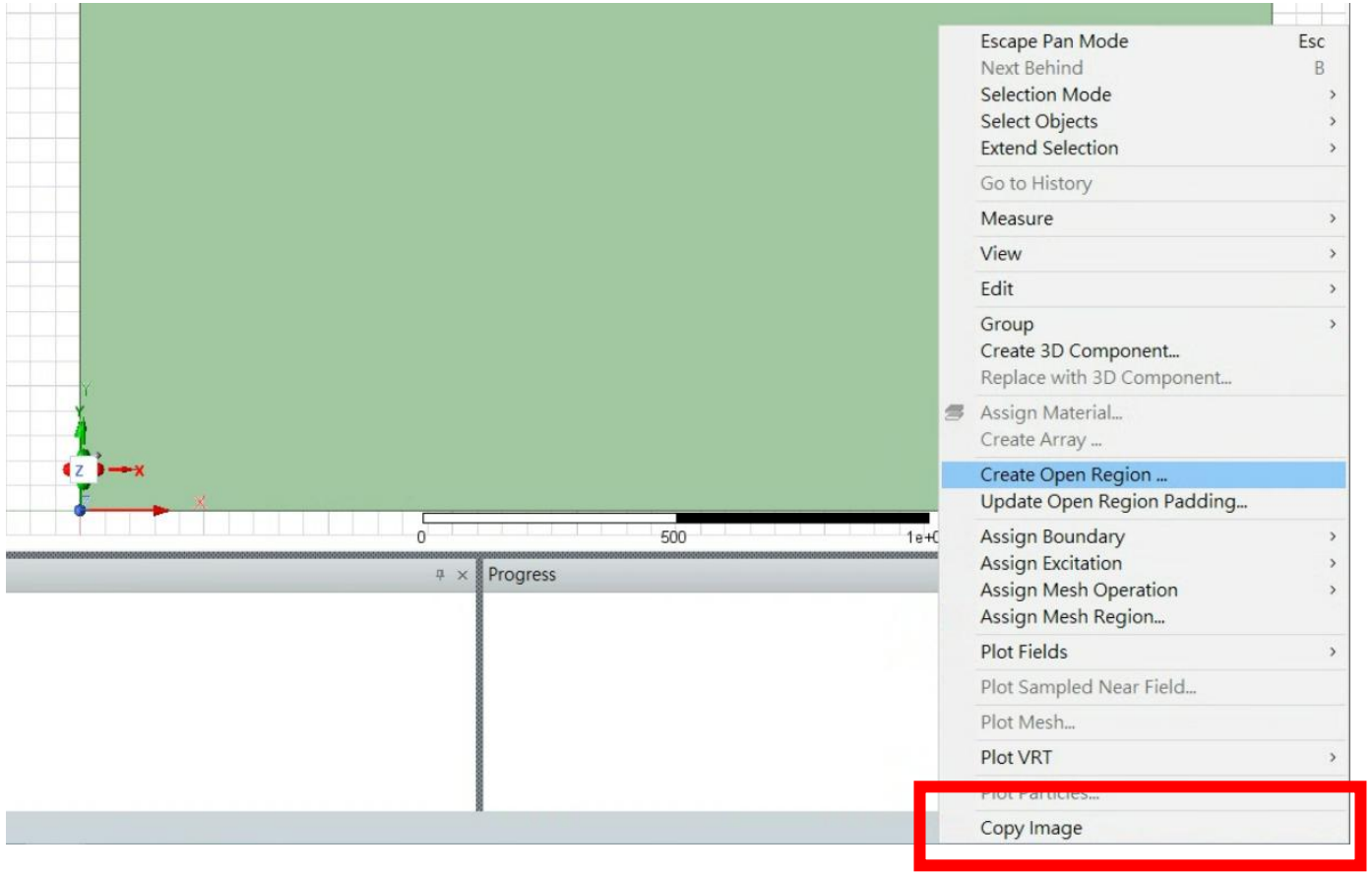


What will you need to export

- Model or result screen shot
- Export data

Model screen shot

- **Right click->copy image**



Lab Training template

Reporter:

Date:

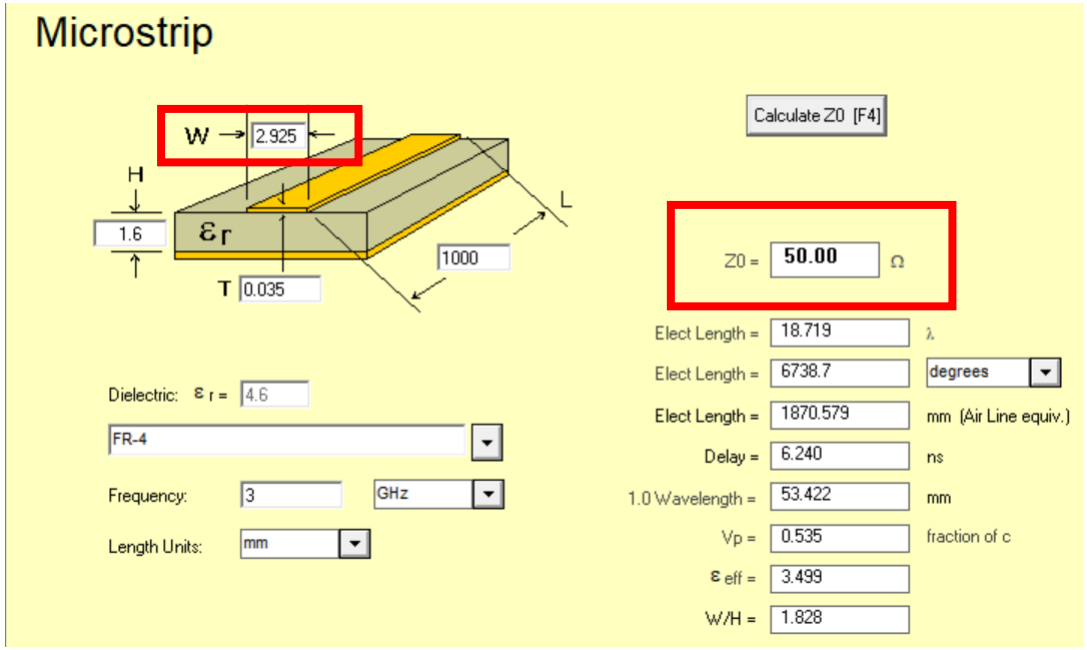
Advisor:

Model parameter calculating

Give out your parameter

$Gap=, Width=, L=, \lambda=$

$, \boldsymbol{f}=$



Model design

substrate	r_{ext}	r_{int}	w	g	s	d_{in}	d_{out}	l
RO4003C	3.8	3.0	0.4	0.4	0.4	5	10	

(θ_D, θ_C)
(30°, 60°)

1-port



Give your figure front and back

Driven model

- Model figure with lumped port
- Lumped port set up
- Analysis set up
- Sweep
- S(2,1) db,phase (big range and small range)
- convergence

eigenmode

- Model figure
- Analysis set up
- convergence
- Eigenmode Q value

conclusion

- table for two simulation result

Driven modal S(2,1)	Peak1		
freq			
dB			

Eigenmode S(2,1)	Peak1		
freq			

Experiment S(2,1)	Peak1		
freq			
dB			

The value or relation between the resonator

Reference

- HFSS help
- Levenson-Falk, E. M. and S. A. Shanto (2025). "A review of design concerns in superconducting quantum circuits." Materials for Quantum Technology **5**(2).
- NAS resource(HFSS training)
- Graduated student source